Homework 4: Rainfall prediction with likelihood kriging

Spatial statistics 2026–2027 · Professor: Dr. Lasinio

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1 My question and data

I want to predict annual rainfall at the supplied grid locations for **each year from 1970 to 1982**. I also want to show how uncertain those predictions are. I fit each year separately: I do not treat repeated measurements at the same station in different years as independent spatial locations.

My main idea is simple. Elevation and location describe the broad rainfall pattern. Kriging then uses the similarity between nearby residuals to adjust that pattern. Maximum likelihood estimates the model parameters, while cross-validation helps me choose the model. This differs from IDW, which uses distance weights without fitting a probability model for the rainfall field.

1.1 Data checks and the matching grid

```
rain_file <- new.env()
invisible(load("data/RainData.RData", envir = rain_file))
rain <- rain_file$dati
grid <- rain_file$grid
years <- sort(unique(rain$anno))
stopifnot(isTRUE(all.equal(grid,
  rbind(rain_file$Basquota32, rain_file$Calabriaquota32),
  check.attributes = FALSE)))

# I correct the labels without removing any station or changing rainfall.
region_fix <- c("4085" = "Basilicata", "4086" = "Basilicata",
                "4367" = "Calabria")
rain$Nome_Regione <- as.character(rain$Nome_Regione)
fix_rows <- rain$ID %in% as.numeric(names(region_fix))
rain$Nome_Regione[fix_rows] <-
  unname(region_fix[as.character(rain$ID[fix_rows])])
stopifnot(identical(rain$totanno, rain_file$dati$totanno),
          nrow(rain) == nrow(rain_file$dati),
          !any(rain$Nome_Regione == "Puglia"))

fields <- c("ID", "anno", "XUTM", "YUTM", "Quota", "totanno")
stopifnot(!anyNA(rain[fields]), !anyNA(grid), all(rain$totanno > 0),
          !anyDuplicated(grid[c("XUTM", "YUTM")]))
for (yr in years) {
  d <- subset(rain, anno == yr)
  stopifnot(!anyDuplicated(d$ID),
            !anyDuplicated(d[c("XUTM", "YUTM")]))
}

# These fixed changes of units also apply to every validation fold.
add_coordinates <- function(d) {
  d$x <- d$XUTM / 1000
  d$y <- d$YUTM / 1000
  d$east <- (d$x - 1100) / 100
  d$north <- (d$y - 4400) / 100
  d$elev <- d$Quota / 1000
  d
}
rain <- add_coordinates(rain)
grid <- add_coordinates(grid)
rain$rain <- rain$totanno
annual <- split(rain, rain$anno)

data_summary <- do.call(rbind, lapply(annual, function(d) {
```

```
  data.frame(Year = d$anno[1], Stations = nrow(d),
    Mean_mm = mean(d$rain), SD_mm = sd(d$rain),
    Min_mm = min(d$rain), Max_mm = max(d$rain),
    Cor_elevation = cor(d$rain, d$Quota))
}))
knitr::kable(data_summary, digits = 2, row.names = FALSE)
```

Year	Stations	Mean_mm	SD_mm	Min_mm	Max_mm	Cor_elevation
1970	189	829.08	317.29	289.82	1992.19	0.43
1971	189	988.65	357.80	334.21	2294.56	0.46
1972	187	1187.11	390.21	484.87	2471.26	0.49
1973	188	1241.49	464.21	464.89	2757.61	0.51
1974	187	973.21	407.16	300.55	2425.81	0.40
1975	182	801.19	248.46	212.56	1493.27	0.42
1976	181	1320.34	454.36	590.70	3048.60	0.33
1977	183	619.38	301.14	185.24	1831.80	0.30
1978	182	1027.72	413.07	240.11	2426.23	0.43
1979	181	1027.06	440.18	189.33	2642.56	0.46
1980	178	1104.70	484.78	211.66	2945.60	0.43
1981	173	820.78	424.09	199.14	2191.01	0.39
1982	173	893.29	359.94	160.00	2144.86	0.33

There are **2373 annual observations**, with no missing values in the variables used here. All three corrected stations remain in the data. The matching land grid has **1005 points** and an elevation at each point. It is already inside `RainData.RData`. The separately supplied `AllGrids.RData` belongs to a marine dataset from other years, so I do not use it for rainfall.

As in Homework 2, I interpret rainfall as **mm**, elevation as **m**, and the supplied projected coordinates as **m**. These units are assumptions based on the names and values; the files do not contain complete metadata. I divide coordinates by 1,000 for distances in km and keep their common projected system. I do not assign an unverified EPSG code. The extra centring and scaling in `east`, `north` and `elev` help numerical fitting; distances still use `x` and `y` in km.

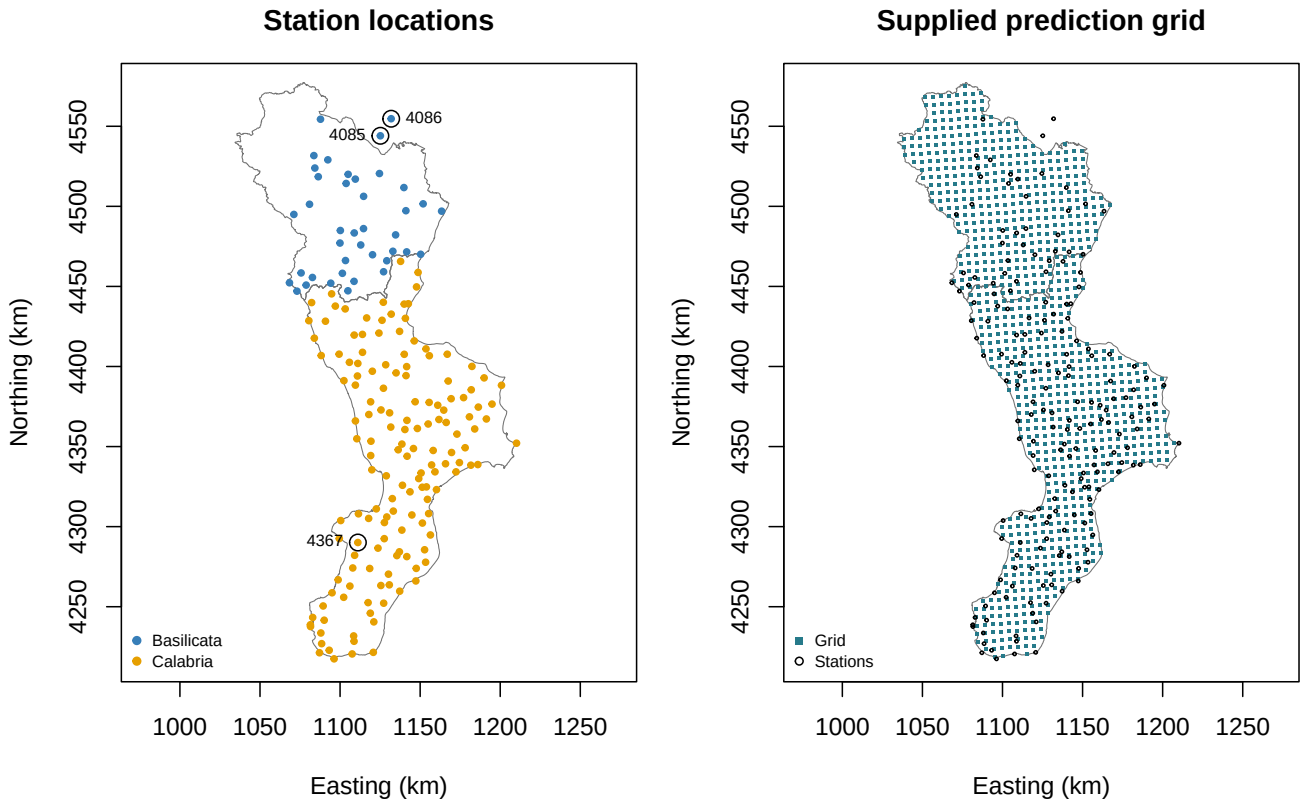
Elevation is the environmental descriptor available at **both** stations and grid points. Latitude and longitude repeat location information, and the grid has no supplied regional factor. I therefore use elevation and the projected coordinates as model covariates. I use the corrected regions to describe the stations, not as an extra predictor.

1.2 Where are the stations?

```
draw_outlines <- function() {
  lines(rain_file$Bas$XUTM/1000, rain_file$Bas$YUTM/1000,
        col = "grey45", lwd = 0.7)
  lines(rain_file$Cal$XUTM/1000, rain_file$Cal$YUTM/1000,
        col = "grey45", lwd = 0.7)
}
map_frame <- function(title) {
  plot(grid$x, grid$y, type = "n", asp = 1,
    xlim = range(c(grid$x, rain$x)), ylim = range(c(grid$y, rain$y)),
    xlab = "Easting (km)", ylab = "Northing (km)", main = title)
  draw_outlines()
}
region_col <- c(Basilicata = "#377EB8", Calabria = "#E69F00")
```

```
stations <- rain[!duplicated(rain$ID), ]
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
```

```
map_frame("Station locations")
points(stations$x, stations$y, pch = 19, cex = 0.55,
       col = region_col[stations$Nome_Regione])
i <- match(as.numeric(names(region_fix)), stations$ID)
points(stations$x[i], stations$y[i], pch = 1, cex = 1.5)
text(stations$x[i], stations$y[i], stations$ID[i],
     pos = c(2, 4, 2), cex = 0.65)
legend("bottomleft", names(region_col), col = region_col,
       pch = 19, bty = "n", cex = 0.7)
map_frame("Supplied prediction grid")
points(grid$x, grid$y, pch = 15, cex = 0.35, col = "#267A8A")
points(stations$x, stations$y, pch = 1, cex = 0.3)
legend("bottomleft", c("Grid", "Stations"), pch = c(15, 1),
       col = c("#267A8A", "black"), bty = "n", cex = 0.7)
```

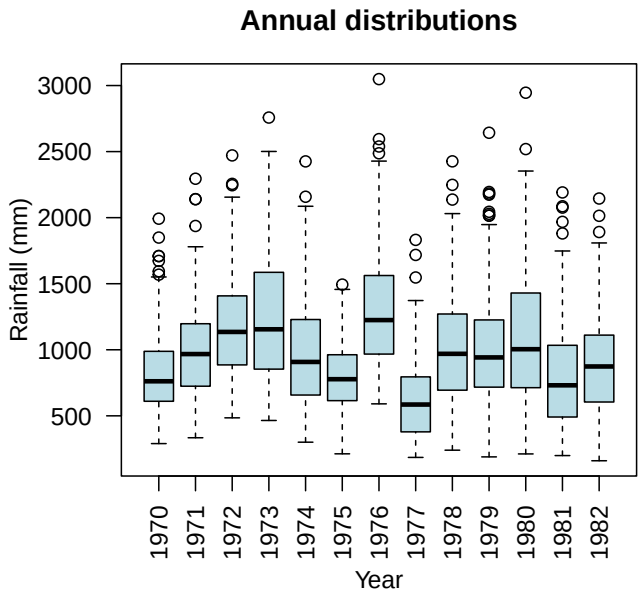


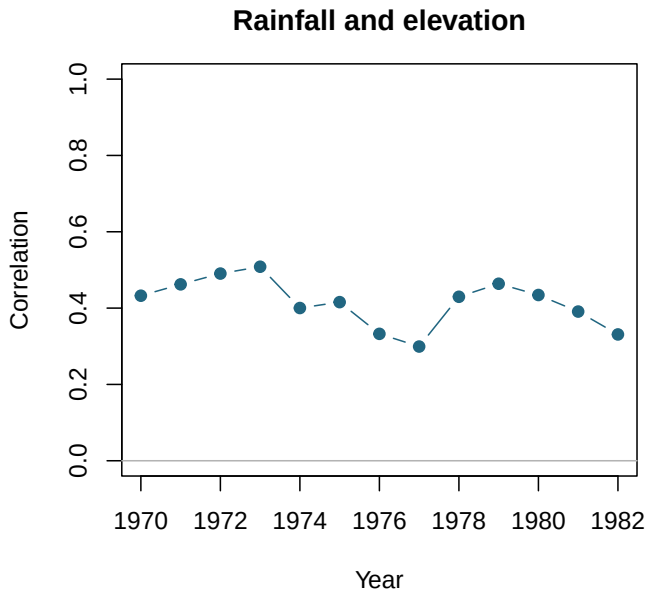
The stations are unevenly spaced along an elongated region. I expect different prediction uncertainty in dense and sparse parts of this region. The number of available stations changes by year, so I use that year's locations when checking the support of each prediction map.

2 Exploratory analysis before fitting

```
par(mfrow = c(1, 2), mar = c(5, 4, 3, 1))
boxplot(rain ~ anno, data = rain, las = 2, col = "#B9DCE5",
        xlab = "Year", ylab = "Rainfall (mm)", main = "Annual distributions")
plot(data_summary$Year, data_summary$Cor_elevation, type = "b",
     ylim = c(0, 1), xlab = "Year", ylab = "Correlation",
     main = "Rainfall and elevation", pch = 19, col = "#216681")
```

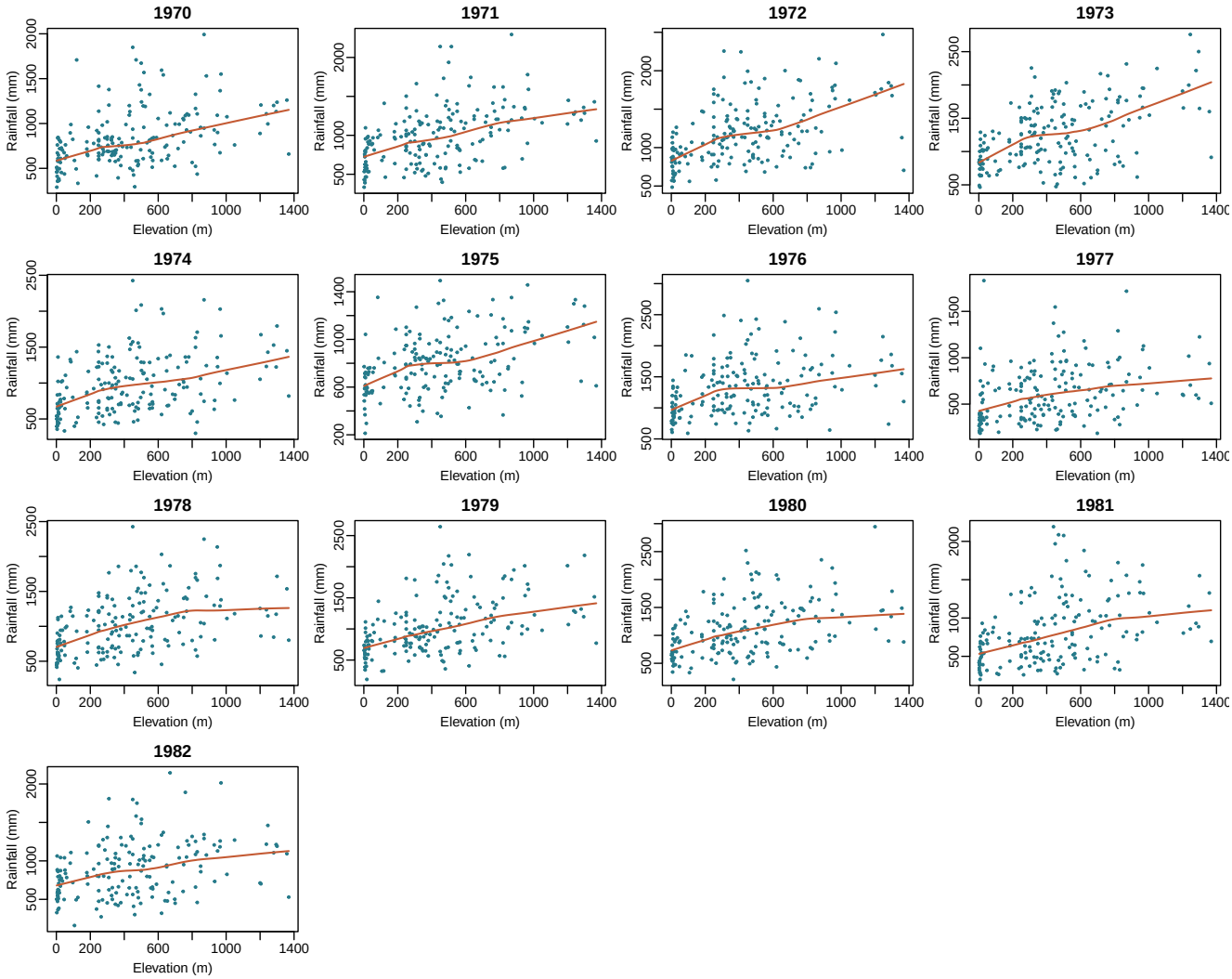
```
abline(h = 0, col = "grey70")
```





Rainfall and elevation have a positive association in every year: the correlations range from **0.30** to **0.51**. This supports including elevation, but does not show a causal effect or explain all local variation. The annual boxplots also show that the rainfall distribution changes with the year. The highest observed station average is in **1976** (**1320.3 mm**), and the lowest is in **1977** (**619.4 mm**). Changes in station coverage mean that these averages are not area-weighted regional rainfall totals or a formal time-trend analysis.

```
par(mfrow = c(4, 4), mar = c(3, 3, 2, 0.5), mgp = c(1.7, 0.5, 0))
for (d in annual) {
  plot(d$Quota, d$rain, pch = 19, cex = 0.4, col = "#267A8A",
       xlab = "Elevation (m)", ylab = "Rainfall (mm)",
       main = as.character(d$anno[1]))
  lines(lowess(d$Quota, d$rain), col = "#C45A33", lwd = 1.5)
}
```



The smooth lines help me see the association without assuming it is exactly linear. I use a linear elevation term as an understandable starting model, and also consider a broad east–north trend. I do not add many polynomial terms just to improve an in-sample fit.

2.1 Distribution, trend and transformation

```
skewness <- function(z) mean((z-mean(z))^3) / mean((z-mean(z))^2)^1.5
distribution_check <- do.call(rbind, lapply(annual, function(d) {
  raw_res <- resid(lm(rain ~ elev + east + north, data = d))
  log_res <- resid(lm(log(rain) ~ elev + east + north, data = d))
  data.frame(Year = d$anno[1], Raw_skew = skewness(d$rain),
    Raw_residual_skew = skewness(raw_res),
    Log_residual_skew = skewness(log_res))
}))
knitr::kable(distribution_check, digits = 2, row.names = FALSE)
```

Year	Raw_skew	Raw_residual_skew	Log_residual_skew
1970	1.05	0.89	-0.10
1971	0.72	0.92	0.04
1972	0.71	0.35	-0.31
1973	0.55	0.26	-0.22

Year	Raw_skew	Raw_residual_skew	Log_residual_skew
1974	0.84	0.55	-0.38
1975	0.41	0.25	-0.80
1976	0.92	0.91	0.14
1977	1.04	0.71	-0.43
1978	0.68	0.57	-0.26
1979	0.94	0.56	-0.58
1980	0.85	0.57	-0.50
1981	0.93	0.78	-0.23
1982	0.67	0.57	-0.63

The rainfall distributions are right-skewed. Removing a trend changes the distribution, and using log rainfall often helps, but not in every year. Skewness alone cannot establish Gaussianity. I therefore compare **raw and log rainfall models** using held-out predictions on the same mm scale, and show Q–Q diagnostics for the chosen models later. For a log model, the transformation is used in fitting, variograms, prediction and intervals; I return to mm only with the appropriate back-transformation.

3 Choosing and fitting the models

3.1 What the models mean

For each year I use the spatial model

$$T\{Y(s)\} = X(s)\beta + w(s) + \epsilon(s).$$

Here, $Y(s)$ is annual rainfall and T is either the identity or the natural logarithm. The trend $X(s)\beta$ describes the broad pattern. The zero-mean process $w(s)$ describes remaining spatial variation, and $\epsilon(s)$ has variance τ^2, the nugget. I assume the two components are independent. The residual covariance is stationary and isotropic: it depends on distance, not on the absolute position or direction.

I compare two trends: **elevation**, and **elevation plus east and north coordinates**. For each trend and response scale, I compare exponential, spherical and Matérn covariance with fixed $\kappa = 1.5$. This gives **12 candidates per year**. The exponential allows a rougher field; the spherical reaches zero correlation at a finite distance; Matérn 1.5 allows a smoother field. Matérn 0.5 would repeat the exponential, so I do not add it as a separate candidate.

All fits use **maximum likelihood**, not REML or Bayesian estimation. For each candidate, I look for the parameter values that make the observed spatial data most plausible under that model. The semivariogram for nonzero distance is $\gamma(h) = \tau^2 + \sigma^2\{1 - \rho(h)\}$, where σ^2 is the partial sill. The total sill is $\tau^2 + \sigma^2$. I estimate the nugget rather than forcing the model to pass through every measured value.

```
trends <- list(Elevation = ~elev,
               Elevation_xy = ~elev + east + north)
candidates <- expand.grid(
  Scale = c("Raw", "Log"), Trend = names(trends),
  Covariance = c("exponential", "spherical", "matern"),
  stringsAsFactors = FALSE)
candidates$Model <- seq_len(nrow(candidates))
knitr::kable(candidates[c("Model", "Scale", "Trend", "Covariance")],
             row.names = FALSE)
```

Model	Scale	Trend	Covariance
1	Raw	Elevation	exponential
2	Log	Elevation	exponential
3	Raw	Elevation_xy	exponential
4	Log	Elevation_xy	exponential

Model	Scale	Trend	Covariance
5	Raw	Elevation	spherical
6	Log	Elevation	spherical
7	Raw	Elevation_xy	spherical
8	Log	Elevation_xy	spherical
9	Raw	Elevation	matern
10	Log	Elevation	matern
11	Raw	Elevation_xy	matern
12	Log	Elevation_xy	matern

3.2 How I compare predictions

I use **five-fold cross-validation within each year**. All candidates use the same split for that year. Each fit uses only the training stations; the held-out rainfall values are used only to score predictions. The random seeds are fixed, and no year contributes observations to another year's model.

Because this homework asks for uncertainty intervals as well as a map, my primary criterion is the **95% interval score**, averaged over all held-out stations:

$$IS_{0.05} = (u - l) + 40(l - y)\,1(y < l) + 40(y - u)\,1(y > u).$$

This is a proper score for the central 95% interval. It penalises intervals that are too wide and intervals that miss the observation. Lower is better. Coverage and average width help explain the score, but coverage by itself is not a proper score: a very wide interval could cover almost everything. I select the lowest mean interval score. I also report **RMSE in mm** for the predictive mean, calculated from all held-out errors together. Squared error is a proper score for a predictive mean, and RMSE has the same ranking as mean squared error. RMSE breaks an exact interval-score tie. This makes interval quality the main objective, while still showing the cost in mean prediction accuracy. I do not change the selection rule between years.

This distinction matters here: a raw Gaussian model can have a slightly smaller RMSE while giving negative lower limits for rainfall. I compare both scales, but I do not treat a tiny RMSE gain as enough evidence of better uncertainty estimates.

These are **model-selection scores**. The best score can be optimistic because I use it to choose among 12 candidates. Unlike the separate nested assessment in Homework 3, I do not claim that this table is an independent assessment of the complete selection procedure. Random folds mainly assess interpolation among existing stations, not prediction across a large unsampled region. A spatially separated outer validation would answer a harder question.

3.3 Reusable fitting and prediction functions

I repeat each likelihood fit from two starting range values and keep the converged fit with the higher likelihood. This reduces dependence on a single starting value, although it cannot prove a global maximum. Failed starts are counted; if neither start converges, the analysis stops rather than silently removing a model or station. I also inspect boundary estimates.

```
design <- function(d, trend) {
  geoR::trend.spatial(trends[[trend]], d)
}

fit_one <- function(d, candidate) {
  z <- if (candidate$Scale == "Log") log(d$rain) else d$rain
  g <- geoR::as.geodata(data.frame(x = d$x, y = d$y, z = z,
    elev = d$elev, east = d$east, north = d$north), covar.col = 4:6)
  X <- design(d, candidate$Trend)
  s2 <- mean(lm.fit(X, z)$residuals^2)
  fits <- lapply(c(20, 80), function(start) {
    tryCatch(geoR::likfit(g, trend = trends[[candidate$Trend]],
```

```
      cov.model = candidate$Covariance, kappa = 1.5, fix.kappa = TRUE,
      ini.cov.pars = c(0.8*s2, start), nugget = 0.2*s2,
      fix.nugget = FALSE, lambda = 1, fix.lambda = TRUE,
      lik.method = "ML", limits = geoR::pars.limits(phi = c(0.5, 1500)),
      messages = FALSE, control = list(maxit = 2000)),
      error = function(e) NULL)
  })
  ok <- vapply(fits, function(m) {
    !is.null(m) && is.finite(m$loglik) &&
      m$info.minimisation.function$convergence == 0
  }, logical(1))
  if (!any(ok)) stop("No converged fit for this candidate.")
  valid <- fits[ok]
  fit <- valid[[which.max(vapply(valid, function(m) m$loglik, numeric(1)))]]
  list(fit = fit, g = g, X = X, rejected = sum(!ok))
}

predict_one <- function(f, locations, candidate) {
  p <- geoR::krige.conv(f$g,
    locations = as.matrix(locations[c("x", "y")]),
    krige = geoR::krige.control(type.krige = "OK", obj.model = f$fit,
      trend.d = f$X, trend.l = design(locations, candidate$Trend),
      micro.scale = 0),
    output = geoR::output.control(signal = FALSE, messages = FALSE))
  mu <- p$predict
  v <- p$krige.var
  stopifnot(all(is.finite(mu)), all(is.finite(v)), all(v > 0))
  lower <- mu - qnorm(0.975)*sqrt(v)
  upper <- mu + qnorm(0.975)*sqrt(v)
  if (candidate$Scale == "Log") {
    answer <- data.frame(Mean = exp(mu + v/2),
                         Lower = exp(lower), Upper = exp(upper))
  } else {
    answer <- data.frame(Mean = mu, Lower = lower, Upper = upper)
  }
  answer$mu <- mu
  answer$variance <- v
  stopifnot(all(is.finite(as.matrix(answer))))
  answer
}

score_predictions <- function(y, p) {
  interval_score <- (p$Upper-p$Lower) +
    40*pmax(p$Lower-y, 0) + 40*pmax(y-p$Upper, 0)
  data.frame(RMSE = sqrt(mean((p$Mean-y)^2)),
    IS95 = mean(interval_score),
    Coverage = 100*mean(y >= p$Lower & y <= p$Upper),
    Width = mean(p$Upper-p$Lower), Bias = mean(p$Mean-y))
}
```

The `OK` option in `geoR` estimates the trend coefficients. With the external elevation and coordinate design matrices above, the prediction is **universal kriging**, despite that option name. I explicitly use `signal = FALSE` to predict an annual observation, including the nugget.

3.4 Pointwise intervals and the log scale

For a raw Gaussian model, the predictive mean is m and the interval is $m \pm 1.96\sqrt{v}$. For a log model, m and v refer to log rainfall. On the original scale I use

$$\widehat{E(Y \mid \text{data})} = \exp(m + v/2), \qquad [l, u] = [\exp(m - 1.96\sqrt{v}),\ \exp(m + 1.96\sqrt{v})].$$

Simply using $\exp(m)$ would give the predictive median, not the mean needed for squared-error comparison. I apply the same formulas in cross-validation and on the grid. I log the data explicitly and keep `lambda = 1` in `geoR` to avoid applying the transformation twice.

For the homework's pointwise confidence intervals, I report **approximate 95% prediction intervals for unknown annual rainfall at each location**. They include spatial uncertainty, the nugget and the universal-kriging correction for estimating the mean coefficients. They treat the fitted covariance parameters and selected model as fixed, so they do not include their uncertainty. They are not confidence intervals for a regression coefficient, Bayesian credible intervals, or a simultaneous 95% band for all grid points. Intervals for only the smooth latent signal would be narrower because they would exclude the new observation's nugget.

```
analyse_year <- function(d, grid) {
  set.seed(2026 + d$anno[1])
  fold <- sample(rep(1:5, length.out = nrow(d)))
  predictions <- vector("list", nrow(candidates))
  scores <- vector("list", nrow(candidates))
  rejected <- integer(nrow(candidates))
  for (j in seq_len(nrow(candidates))) {
    p <- data.frame(Mean = rep(NA_real_, nrow(d)), Lower = NA_real_,
                    Upper = NA_real_, mu = NA_real_, variance = NA_real_)
    for (k in 1:5) {
      test <- fold == k
      f <- fit_one(d[!test, ], candidates[j, ])
      p[test, ] <- predict_one(f, d[test, ], candidates[j, ])
      rejected[j] <- rejected[j] + f$rejected
    }
    predictions[[j]] <- p
    scores[[j]] <- score_predictions(d$rain, p)
  }
  scores <- cbind(candidates, do.call(rbind, scores))
  ranking <- order(scores$IS95, scores$RMSE)
  best <- ranking[1]
  fitted <- fit_one(d, candidates[best, ])
  mapped <- predict_one(fitted, grid, candidates[best, ])
  hull <- chull(d$x, d$y)
  mapped$Outside_hull <- sp::point.in.polygon(
    grid$x, grid$y, d$x[hull], d$y[hull]) == 0
  mapped$Outside_elevation <- grid$Quota < min(d$Quota) |
                              grid$Quota > max(d$Quota)
  list(year = d$anno[1], fold = fold, scores = scores,
    predictions = predictions, best = best, runner_up = ranking[2],
    fitted = fitted, mapped = mapped,
    rejected = sum(rejected) + fitted$rejected)
}
```

```
# I repeat the same independent yearly analysis; seeds are set inside it.
if (.Platform$OS.type == "unix") {
  results <- parallel::mclapply(annual, analyse_year, grid = grid,
    mc.cores = 4, mc.set.seed = FALSE)
} else {
  results <- lapply(annual, analyse_year, grid = grid)
```

```
}
stopifnot(!any(vapply(results, inherits, logical(1), "try-error")))
```

4 Results and model choices

```
selected <- do.call(rbind, lapply(results, function(r) {
  a <- r$scores[r$best, ]
  data.frame(Year = r$year, Model = a$Model, Scale = a$Scale,
    Trend = a$Trend, Covariance = a$Covariance, IS95_mm = a$IS95,
    RMSE_mm = a$RMSE)
}))
knitr::kable(selected, digits = 2, row.names = FALSE)
```

Year	Model	Scale	Trend	Covariance	IS95_mm	RMSE_mm
1970	6	Log	Elevation	spherical	1009.55	195.42
1971	6	Log	Elevation	spherical	1020.53	204.66
1972	6	Log	Elevation	spherical	1312.81	257.13
1973	4	Log	Elevation_xy	exponential	1382.17	291.85
1974	10	Log	Elevation	matern	1233.84	255.59
1975	10	Log	Elevation	matern	1003.12	192.90
1976	2	Log	Elevation	exponential	1576.34	334.48
1977	12	Log	Elevation_xy	matern	908.78	182.30
1978	4	Log	Elevation_xy	exponential	1355.00	253.33
1979	4	Log	Elevation_xy	exponential	1422.80	266.25
1980	8	Log	Elevation_xy	spherical	1829.40	333.74
1981	2	Log	Elevation	exponential	1340.30	267.35
1982	2	Log	Elevation	exponential	1316.53	243.52

The selected model has the smallest mean interval score **among these candidates on this split**. It is not a uniquely established model. The complete candidate scores are in the appendix, so the choice can be checked rather than taken on trust. Small differences between candidates should not receive a strong scientific interpretation.

I select log models in **13 of 13 years**, and the elevation-plus-coordinate trend in **5 years**. The choices can change with the validation split or the candidate list. The winning interval-score models use the log scale in every year; this is a result of the comparison, not a requirement imposed on the candidates.

```
interval_results <- do.call(rbind, lapply(results, function(r) {
  a <- r$scores[r$best, ]
  data.frame(Year = r$year, IS95_mm = a$IS95,
    Coverage_percent = a$Coverage, Width_mm = a$Width, Bias_mm = a$Bias,
    RMSE_cost_mm = a$RMSE - min(r$scores$RMSE))
}))
knitr::kable(interval_results, digits = 2, row.names = FALSE)
```

Year	IS95_mm	Coverage_percent	Width_mm	Bias_mm	RMSE_cost_mm
1970	1009.55	92.59	663.77	4.75	0.00
1971	1020.53	91.53	665.68	3.24	2.34
1972	1312.81	94.65	973.76	8.00	1.46
1973	1382.17	95.74	1103.43	8.46	3.92
1974	1233.84	95.72	954.96	13.97	2.86
1975	1003.12	92.86	798.05	4.51	1.28
1976	1576.34	95.03	1235.90	7.59	0.00
1977	908.78	92.90	668.21	-0.13	0.59

Year	IS95_mm	Coverage_percent	Width_mm	Bias_mm	RMSE_cost_mm
1978	1355.00	95.05	1022.90	11.71	2.11
1979	1422.80	93.92	1083.03	-0.83	7.87
1980	1829.40	94.94	1346.19	8.80	3.76
1981	1340.30	94.22	1019.16	6.60	0.63
1982	1316.53	94.22	1058.31	11.86	3.73

The selected models have coverage between **91.5%** and **95.7%**, compared with the nominal 95%. The weakest coverage is in **1971**, so choosing by interval score does not make the intervals perfectly calibrated. Positive bias means overprediction. `RMSE_cost_mm` is the extra RMSE compared with the best mean-prediction candidate. Its maximum is **7.87 mm**. This makes the trade-off visible instead of claiming one model is best for every purpose. These diagnostics also use the selection folds, so I do not describe them as independent guarantees of coverage.

```
example <- results[["1981"]]
point_best <- which.min(example$scores$RMSE)
point_score <- example$scores[point_best, ]
interval_score <- example$scores[example$best, ]
negative_limits <- sum(example$predictions[[point_best]]$Lower < 0)
```

For example, in 1981 the raw model's RMSE is **266.73 mm**, compared with **267.35 mm** for the selected log model. However, its interval score is **1612.91 mm**, compared with **1340.30 mm** for the log model. The raw model also gives negative lower limits at **35 held-out stations**. Here I accept a small increase in RMSE for better interval predictions and a distribution that respects the positive rainfall scale.

4.1 Fitted covariance parameters

```
parameters <- do.call(rbind, lapply(results, function(r) {
  m <- r$fitted$fit
  data.frame(Year = r$year, Partial_sill = m$cov.pars[1],
    Nugget = m$nugget, Nugget_share = m$nugget/sum(m$cov.pars[1], m$nugget),
    Phi_km = m$cov.pars[2],
    Range_bound = m$cov.pars[2] < 0.51 | m$cov.pars[2] > 1499,
    Rejected_starts = r$rejected)
}))
knitr::kable(parameters, digits = 3, row.names = FALSE)
```

Year	Partial_sill	Nugget	Nugget_share	Phi_km	Range_bound	Rejected_starts
1970	0.136	0.019	0.124	108.874	FALSE	0
1971	0.091	0.014	0.137	107.022	FALSE	1
1972	0.058	0.019	0.249	41.530	FALSE	1
1973	0.106	0.019	0.153	35.837	FALSE	1
1974	0.146	0.047	0.243	27.512	FALSE	1
1975	0.045	0.050	0.525	16.191	FALSE	0
1976	0.080	0.031	0.276	39.453	FALSE	0
1977	0.203	0.057	0.219	26.045	FALSE	0
1978	0.105	0.034	0.246	47.844	FALSE	1
1979	0.143	0.043	0.233	68.659	FALSE	0
1980	0.119	0.065	0.354	108.174	FALSE	0
1981	0.240	0.055	0.187	77.702	FALSE	2
1982	0.149	0.057	0.277	68.200	FALSE	0

All selected models use log rainfall, so the variance parameters in this table are in **squared log units**. The distance units are unchanged. `Phi_km` is the software's range parameter: it is the exact range for the spherical model, while

an exponential correlation falls to about 0.05 at 3ϕ. For Matérn 1.5, that distance is about 4.74ϕ. I therefore interpret the range using the selected family.

I balance trend fit with residual structure. A smaller nugget is not a selection rule by itself, and a larger nugget does not automatically mean bad measurements. Boundary estimates and noisy diagnostic curves make precise parameter interpretations less reliable. There are **0 selected fits at a range bound** and **7 rejected starts** across the fitting and validation runs. Every candidate still has a converged fit in every fold.

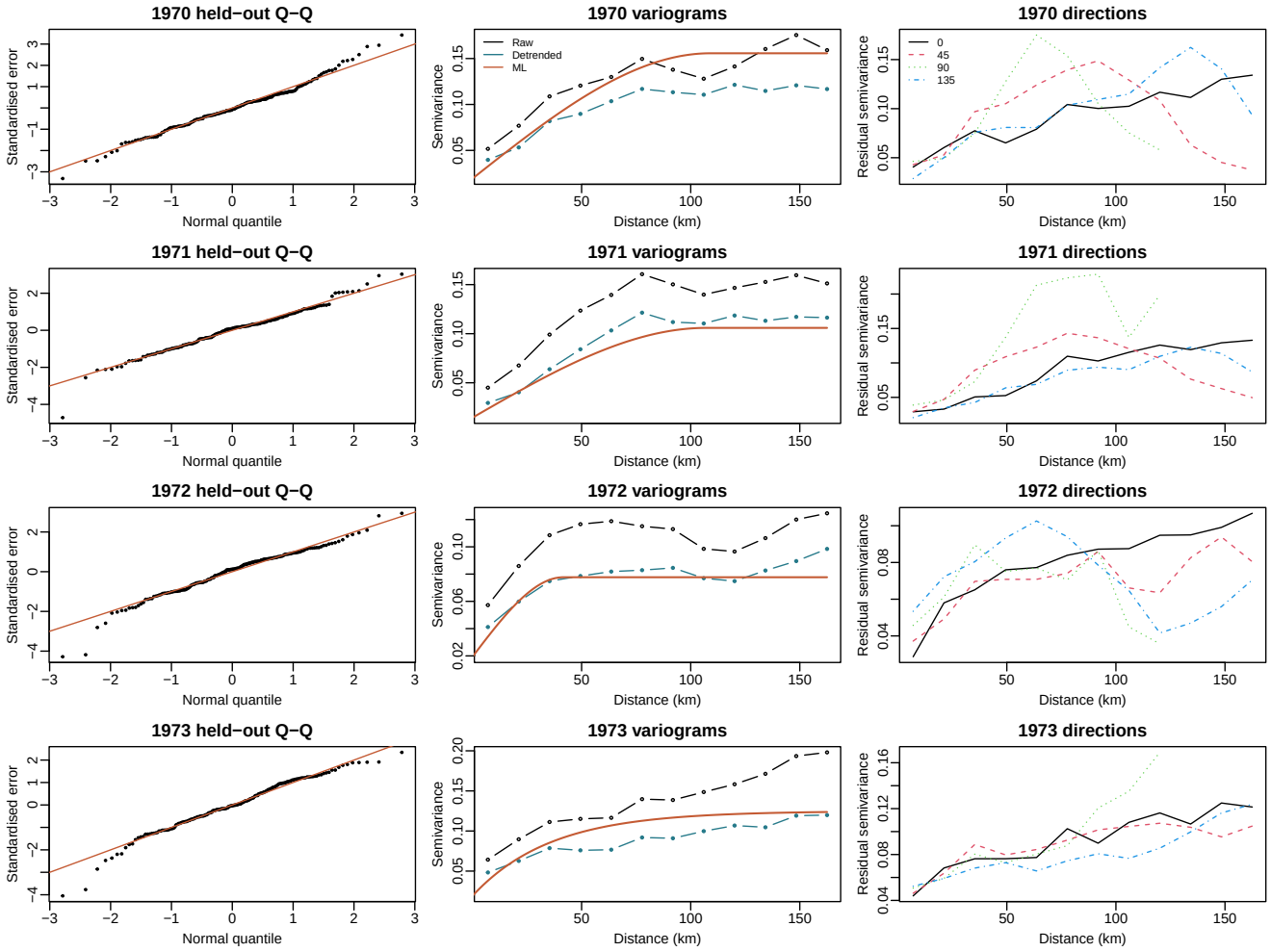
5 Diagnostics for the selected models

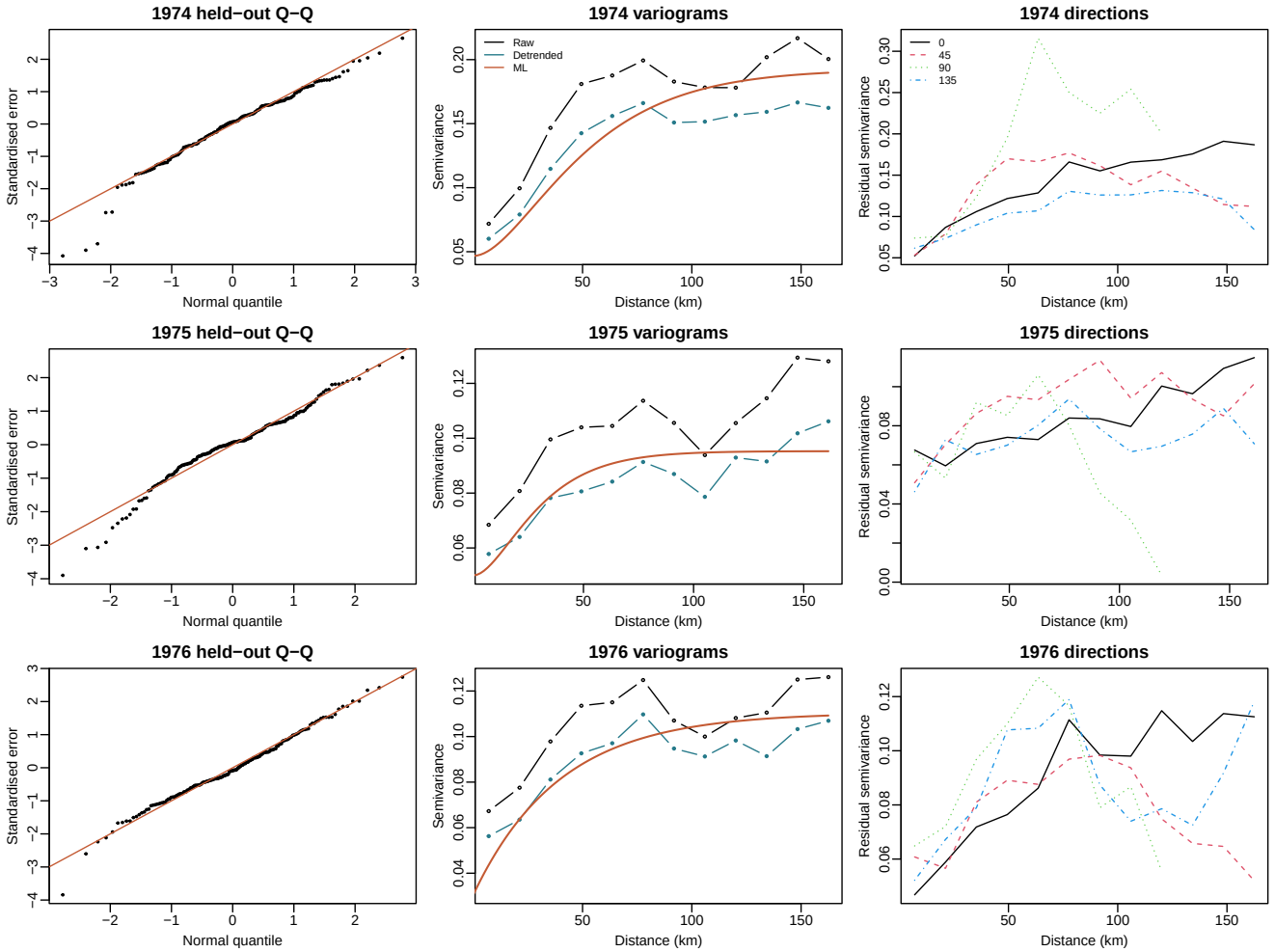
I examine the raw and detrended semivariograms on the **selected response scale**. A rising raw curve can reflect a changing mean, not only residual dependence. The fitted curve comes from maximum likelihood; it is not a least-squares fit to the plotted variogram points. I use 12 bins up to half the maximum station separation. The pair counts are retained with the diagnostics because sparse bins are less stable.

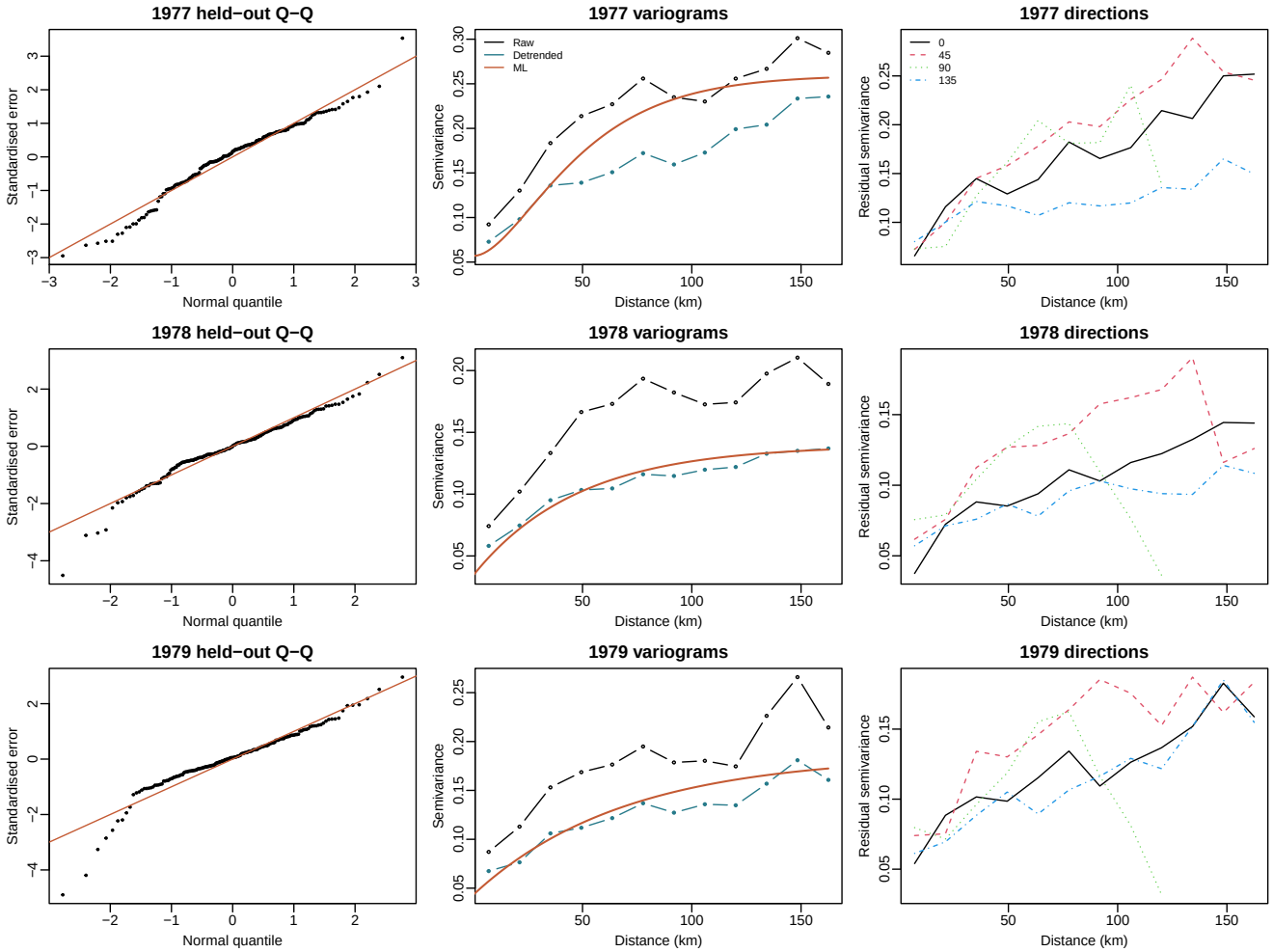
```
diagnostics <- lapply(results, function(r) {
  f <- r$fitted
  cnd <- candidates[r$best, ]
  cutoff <- max(dist(f$g$coords))/2
  breaks <- seq(0, cutoff, length.out = 13)
  raw <- geoR::variog(f$g, breaks = breaks, trend = "cte",
                      messages = FALSE)
  adjusted <- geoR::variog(f$g, breaks = breaks,
    trend = trends[[cnd$Trend]], messages = FALSE)
  dirs <- lapply(c(0, 45, 90, 135), function(a) {
    geoR::variog(f$g, breaks = breaks, trend = trends[[cnd$Trend]],
      direction = a*pi/180, tolerance = pi/8, messages = FALSE)
  })
  list(raw = raw, adjusted = adjusted, directions = dirs)
})
```

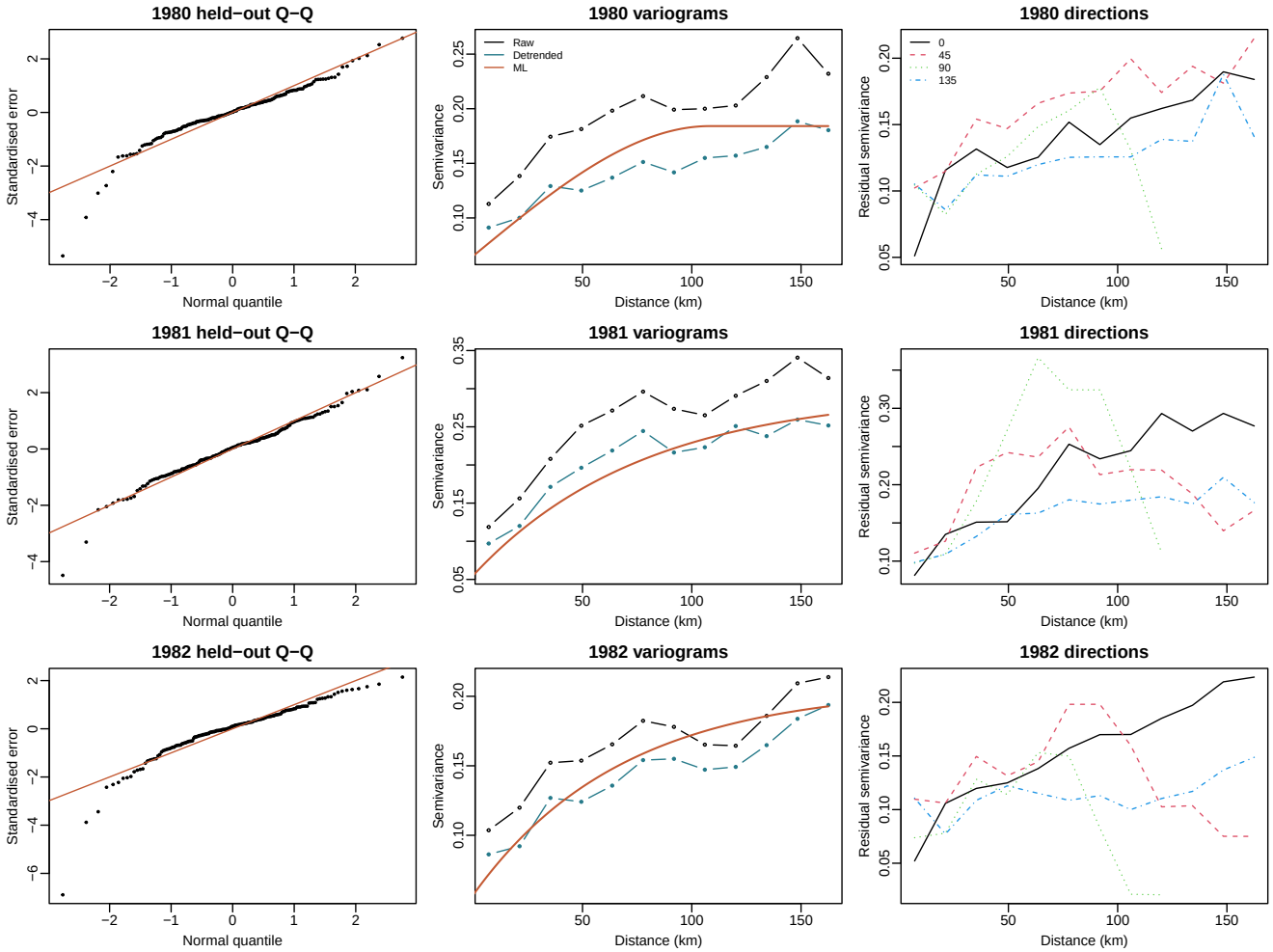
```
groups <- split(seq_along(results),
                cut(seq_along(results), breaks = 4, labels = FALSE))
for (indices in groups) {
  par(mfrow = c(length(indices), 3), mar = c(3, 3, 2, 0.5),
      mgp = c(1.7, 0.5, 0))
  for (i in indices) {
    r <- results[[i]]
    dg <- diagnostics[[i]]
    m <- r$fitted$fit
    cnd <- candidates[r$best, ]
    d <- annual[[i]]
    p <- r$predictions[[r$best]]
    z <- if (cnd$Scale == "Log") log(d$rain) else d$rain
    qqnorm((z-p$mu)/sqrt(p$variance), pch = 19, cex = 0.35,
      main = paste(r$year, "held-out Q-Q"), xlab = "Normal quantile",
      ylab = "Standardised error")
    abline(0, 1, col = "#C45A33")
    h <- seq(0, max(dg$raw$u), length.out = 100)
    curve <- m$nugget + m$cov.pars[1] -
      geoR::cov.spatial(h, cov.model = m$cov.model,
                        cov.pars = m$cov.pars, kappa = m$kappa)
    plot(dg$raw$u, dg$raw$v, type = "b", pch = 1, cex = 0.45,
      ylim = range(c(dg$raw$v, dg$adjusted$v, curve)),
      main = paste(r$year, "variograms"), xlab = "Distance (km)",
      ylab = "Semivariance")
    lines(dg$adjusted$u, dg$adjusted$v, type = "b", pch = 19,
```

```
          cex = 0.45, col = "#267A8A")
    lines(h, curve, col = "#C45A33", lwd = 1.5)
    if (i == indices[1]) legend("topleft", c("Raw", "Detrended", "ML"),
      col = c("black", "#267A8A", "#C45A33"), lty = 1,
      cex = 0.75, bty = "n")
    plot(dg$adjusted$u, dg$adjusted$v, type = "n",
      ylim = range(unlist(lapply(dg$directions, function(v) v$v))),
      main = paste(r$year, "directions"), xlab = "Distance (km)",
      ylab = "Residual semivariance")
    for (k in 1:4) lines(dg$directions[[k]]$u, dg$directions[[k]]$v,
                         col = k, lty = k)
    if (i == indices[1]) legend("topleft", c("0", "45", "90", "135"),
                          col = 1:4, lty = 1:4, cex = 0.75, bty = "n")
  }
  cat("\n\n")
}
```




1974 held−out Q−Q
Standardised error
Normal quantile
1974 variograms
Raw
Detrended
ML
Semivariance
Distance (km)
1974 directions
0
45
90
135
Residual semivariance
Distance (km)
1975 held−out Q−Q
Standardised error
Normal quantile
1975 variograms
Semivariance
Distance (km)
1975 directions
Residual semivariance
Distance (km)
1976 held−out Q−Q
Standardised error
Normal quantile
1976 variograms
Semivariance
Distance (km)
1976 directions
Residual semivariance
Distance (km)


1977 held−out Q−Q
Standardised error
Normal quantile
1977 variograms
Raw
Detrended
ML
Semivariance
Distance (km)
1977 directions
0
45
90
135
Residual semivariance
Distance (km)
1978 held−out Q−Q
Standardised error
Normal quantile
1978 variograms
Semivariance
Distance (km)
1978 directions
Residual semivariance
Distance (km)
1979 held−out Q−Q
Standardised error
Normal quantile
1979 variograms
Semivariance
Distance (km)
1979 directions
Residual semivariance
Distance (km)



The Q–Q plots assess held-out errors on the model scale, including their predicted standard deviations. Tail departures warn me that Gaussian intervals may be imperfect. These errors share training observations, so they are not independent observations for a formal normality test. The directional variograms (angles in degrees) help check the isotropy assumption. Differences can also arise from unequal pair counts, the elongated sampling region, or a trend that is too simple. The plots do not prove stationarity or isotropy. In particular, the lower tails in 1980 and 1982 depart from the normal line. I keep those observations, but unusual dry locations may be harder to predict than the Gaussian approximation suggests. The directional curves also differ at longer distances, so isotropy remains a working approximation. I keep these limits explicit when interpreting the maps.

```
pair_check <- do.call(rbind, lapply(seq_along(results), function(i) {
  dg <- diagnostics[[i]]
  data.frame(Year = results[[i]]$year,
    Min_pairs = min(dg$adjusted$n), Max_pairs = max(dg$adjusted$n),
    Min_directional_pairs = min(unlist(lapply(dg$directions,
                                              function(v) v$n))))
}))
knitr::kable(pair_check, row.names = FALSE)
```

Year	Min_pairs	Max_pairs	Min_directional_pairs
1970	394	1648	7
1971	394	1648	7
1972	388	1614	7
1973	388	1620	7
1974	384	1603	7

Year	Min_pairs	Max_pairs	Min_directional_pairs
1975	355	1513	6
1976	356	1518	7
1977	364	1565	7
1978	360	1542	7
1979	360	1541	7
1980	347	1510	7
1981	333	1408	7
1982	330	1415	6

6 Rainfall maps and pointwise intervals

Each yearly page shows the predicted **mean annual observation**, the lower and upper 95% limits, and the interval width. The first three maps share a colour scale within that year, so their colours are directly comparable. The width map has its own scale. Scales are printed on every page and may change between years. I keep the supplied 1,005 grid points; I do not fill in a different regular grid.

Small black dots are that year's observed stations. A cross marks a grid point outside the station convex hull, and an open triangle marks an elevation outside the range observed that year. I keep these points, but their predictions depend more strongly on model extrapolation. Neither flag captures every possible form of extrapolation.

```
map_summary <- do.call(rbind, lapply(results, function(r) {
  p <- r$mapped
  data.frame(Year = r$year, Min_mean = min(p$Mean), Max_mean = max(p$Mean),
    Median_width = median(p$Upper-p$Lower),
    Outside_hull = sum(p$Outside_hull),
    Outside_elevation = sum(p$Outside_elevation),
    Negative_lower = sum(p$Lower < 0))
}))
knitr::kable(map_summary, digits = 1, row.names = FALSE)
```

Year	Min_mean	Max_mean	Median_width	Outside_hull	Outside_elevation	Negative_lower
1970	316.3	2020.8	615.0	117	57	0
1971	379.5	2314.6	622.4	117	57	0
1972	593.5	2611.4	926.3	117	57	0
1973	495.8	2896.7	936.6	117	57	0
1974	408.5	2289.8	883.4	117	57	0
1975	452.4	1495.2	778.3	119	57	0
1976	726.7	2725.0	1143.9	117	57	0
1977	237.9	1579.6	649.9	117	57	0
1978	414.1	2464.7	934.2	117	57	0
1979	393.8	2849.6	986.5	117	57	0
1980	490.4	2966.2	1203.5	117	57	0
1981	286.3	2380.1	889.0	126	57	0
1982	387.3	2456.2	976.8	127	57	0

The grid elevations range from **0 to 1995 m**, while station elevations across all years range from **2 to 1369 m**. Predictions high in the mountains therefore extend beyond the elevation information in the data. A wide interval is useful evidence of uncertainty, but it may still understate uncertainty from extrapolating an incorrect trend. All the selected models use log rainfall, so their lower limits stay positive. The raw Gaussian candidates can produce negative lower limits, as the 1981 validation example shows. I do not repair such limits by silently truncating them, because that would change the stated model.

For a log model, a large interval width in mm can reflect both uncertainty on the log scale and a high rainfall level. Even the same log-scale standard error gives a wider interval in mm at a wetter location. I therefore do not read

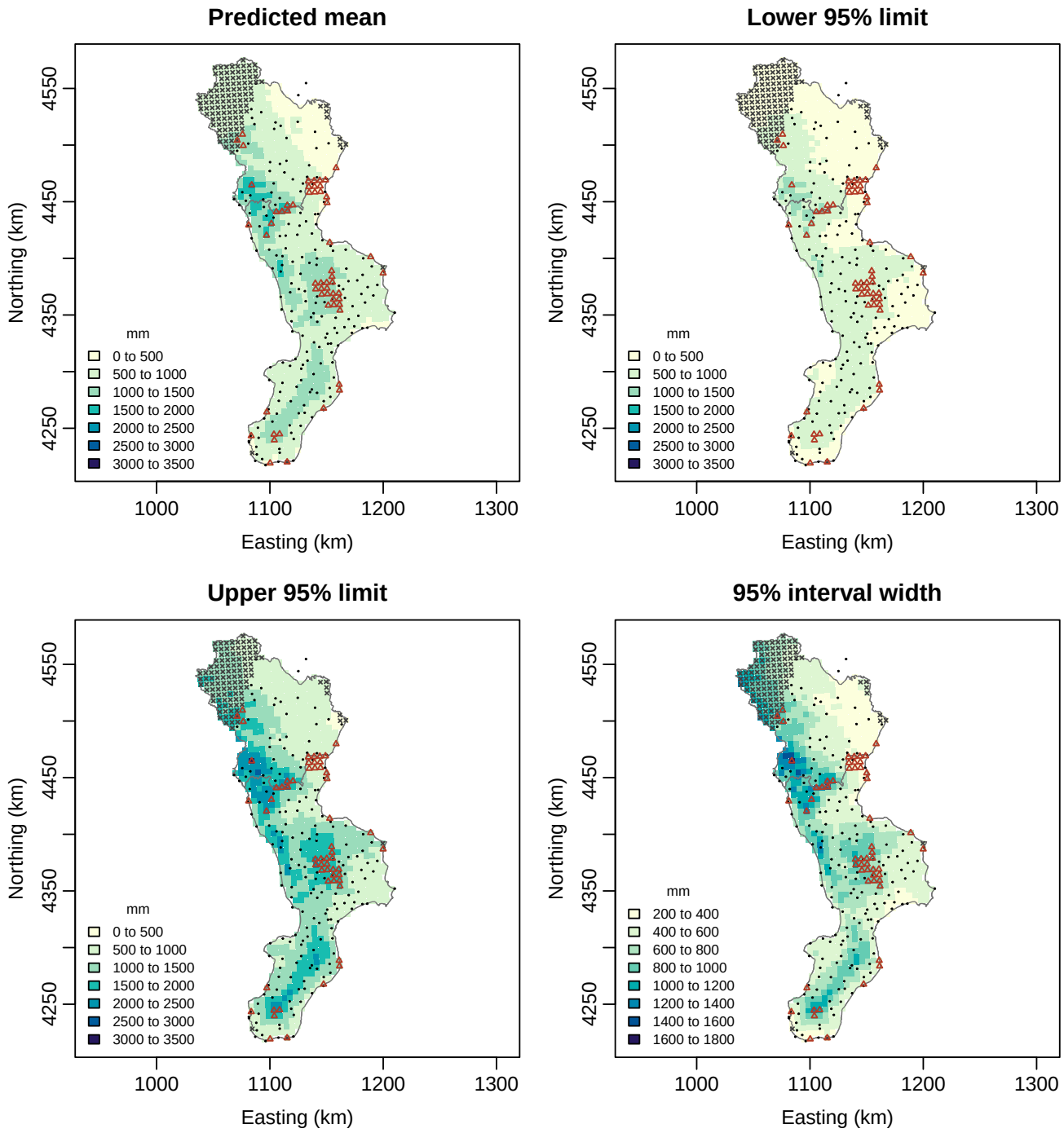
the width map as a map of station density alone.

```
draw_values <- function(values, breaks, title, d, p) {
  colours <- hcl.colors(length(breaks)-1, "YlGnBu", rev = TRUE)
  map_frame(title)
  index <- cut(values, breaks, include.lowest = TRUE)
  points(grid$x, grid$y, pch = 15, cex = 0.55, col = colours[index])
  draw_outlines()
  points(d$x, d$y, pch = 20, cex = 0.18)
  points(grid$x[p$Outside_hull], grid$y[p$Outside_hull],
         pch = 4, cex = 0.38, col = "grey25")
  points(grid$x[p$Outside_elevation], grid$y[p$Outside_elevation],
         pch = 2, cex = 0.48, col = "#B33A26")
  labels <- paste(head(breaks, -1), tail(breaks, -1), sep = " to ")
  legend("bottomleft", labels, fill = colours, title = "mm",
         bty = "n", cex = 0.75)
}
```

```
for (i in seq_along(results)) {
  r <- results[[i]]
  p <- r$mapped
  d <- annual[[i]]
  a <- r$scores[r$best, ]
  cat("\n\n\\newpage\n\n## Year ", r$year, "\n\n", sep = "")
  cat("I use model ", a$Model, ": ", a$Scale, ", ", a$Trend,
      ", ", a$Covariance, ". Selection interval score is ", fmt(a$IS95),
      " mm and RMSE is ", fmt(a$RMSE),
      " mm; held-out coverage is ", fmt(a$Coverage), "%. ",
      "Predicted means range from ", fmt(min(p$Mean)), " to ",
      fmt(max(p$Mean)), " mm. The median interval width is ",
      fmt(median(p$Upper-p$Lower)), " mm.\n\n", sep = "")
  breaks <- pretty(range(p[c("Mean", "Lower", "Upper")]), n = 6)
  width_breaks <- pretty(range(p$Upper-p$Lower), n = 6)
  par(mfrow = c(2, 2), mar = c(3.5, 3.5, 2.3, 0.5),
      mgp = c(2, 0.6, 0))
  draw_values(p$Mean, breaks, "Predicted mean", d, p)
  draw_values(p$Lower, breaks, "Lower 95% limit", d, p)
  draw_values(p$Upper, breaks, "Upper 95% limit", d, p)
  draw_values(p$Upper-p$Lower, width_breaks, "95% interval width", d, p)
  cat("\n\nCrosses: outside station hull. Red triangles: elevation ",
      "outside the observed range. ", sum(p$Outside_hull),
      " grid points are outside the hull and ", sum(p$Outside_elevation),
      " have an extrapolated elevation. These groups may overlap.\n\n",
      sep = "")
}
```

6.1 Year 1970

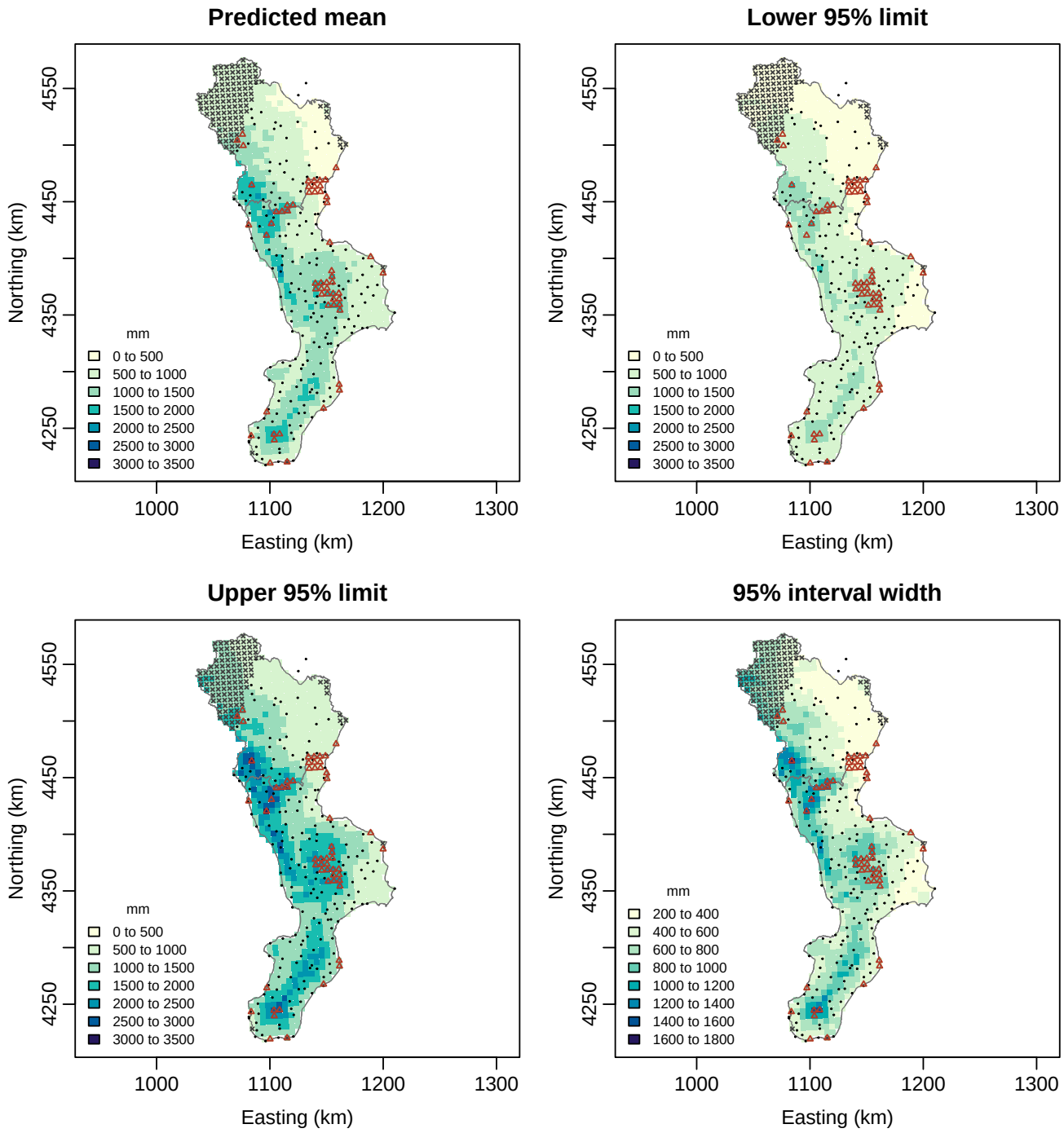
I use model 6: Log, Elevation, spherical. Selection interval score is 1009.6 mm and RMSE is 195.4 mm; held-out coverage is 92.6%. Predicted means range from 316.3 to 2020.8 mm. The median interval width is 615.0 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.2 Year 1971

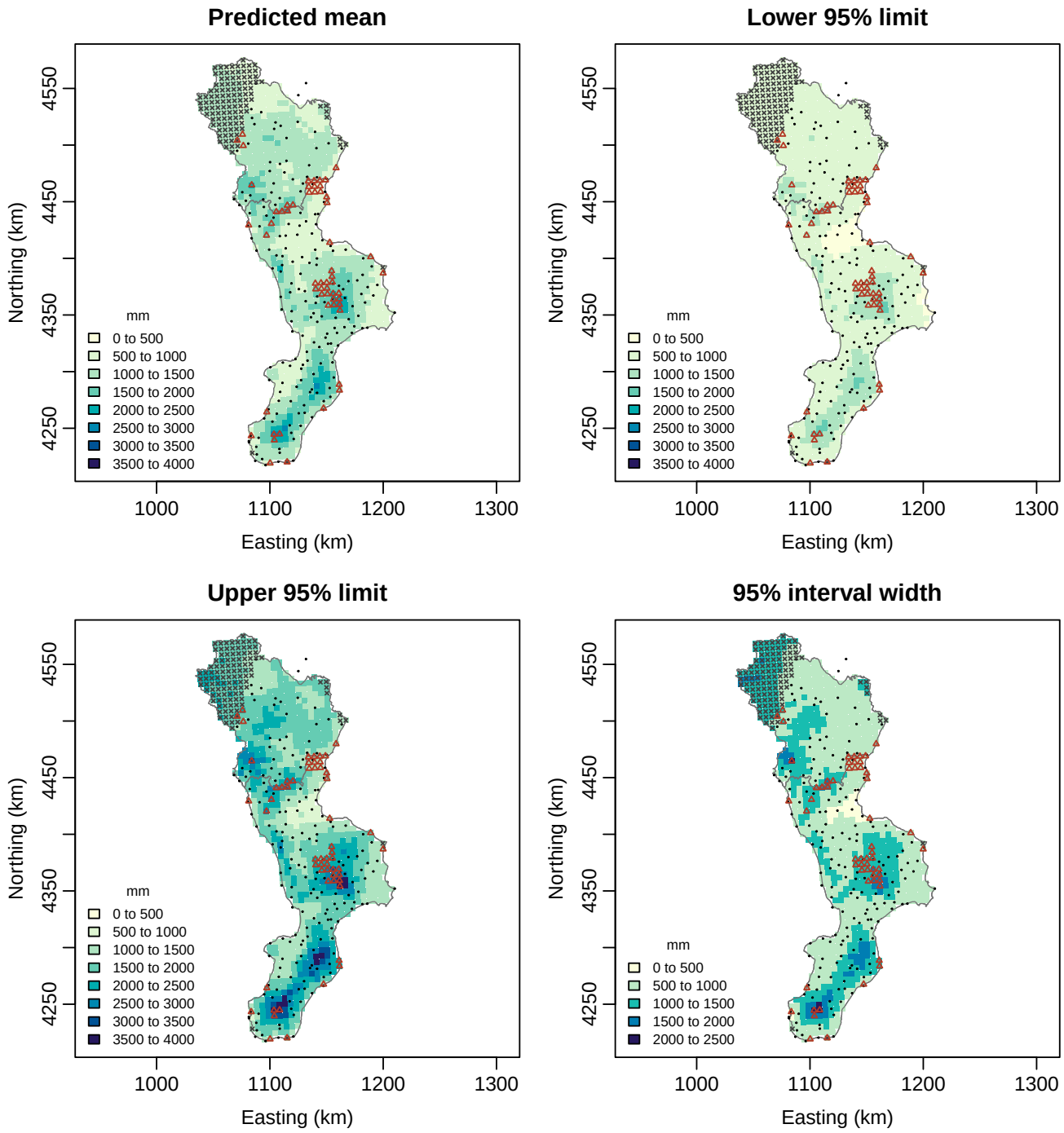
I use model 6: Log, Elevation, spherical. Selection interval score is 1020.5 mm and RMSE is 204.7 mm; held-out coverage is 91.5%. Predicted means range from 379.5 to 2314.6 mm. The median interval width is 622.4 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.3 Year 1972

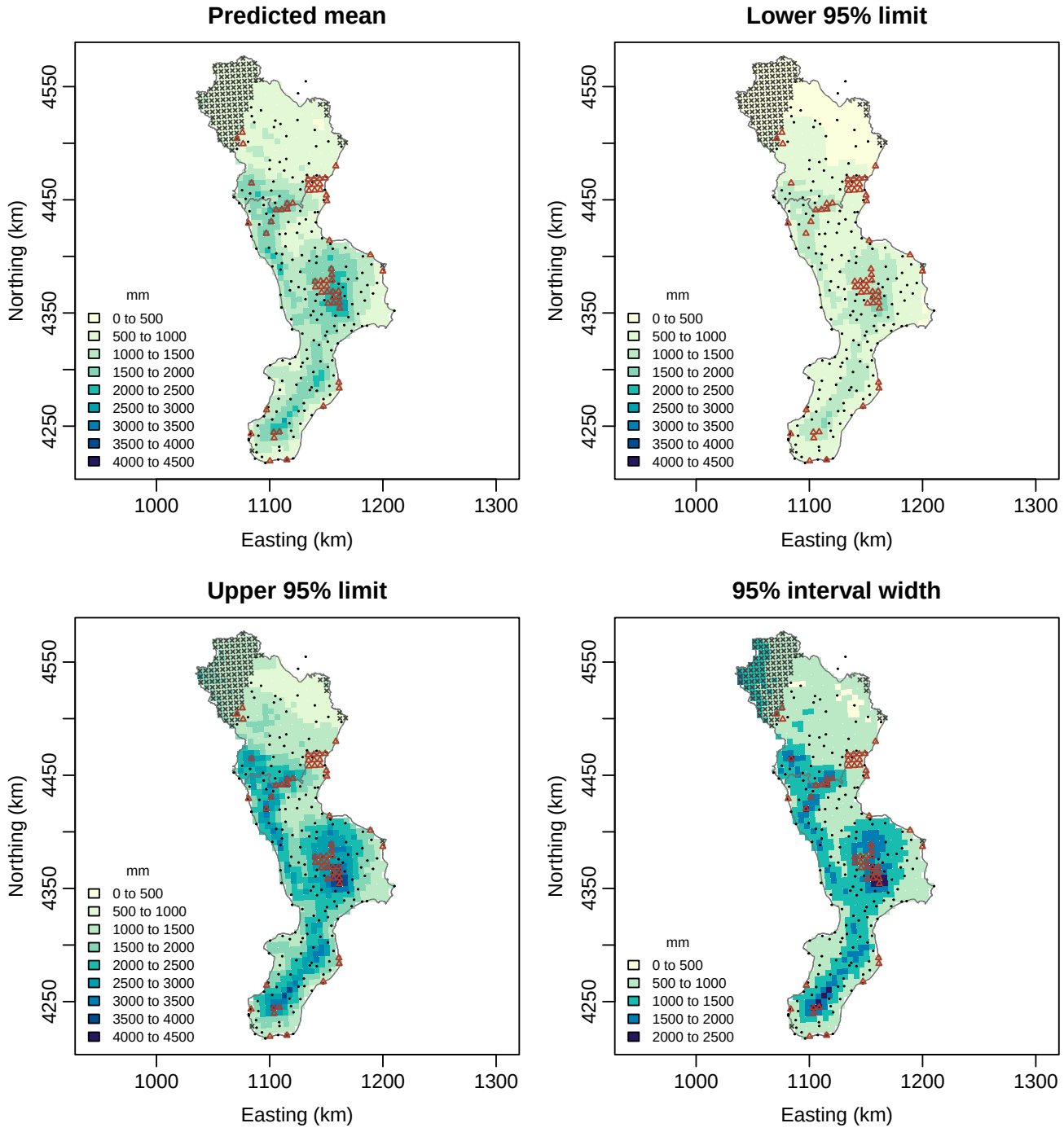
I use model 6: Log, Elevation, spherical. Selection interval score is 1312.8 mm and RMSE is 257.1 mm; held-out coverage is 94.7%. Predicted means range from 593.5 to 2611.4 mm. The median interval width is 926.3 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.4 Year 1973

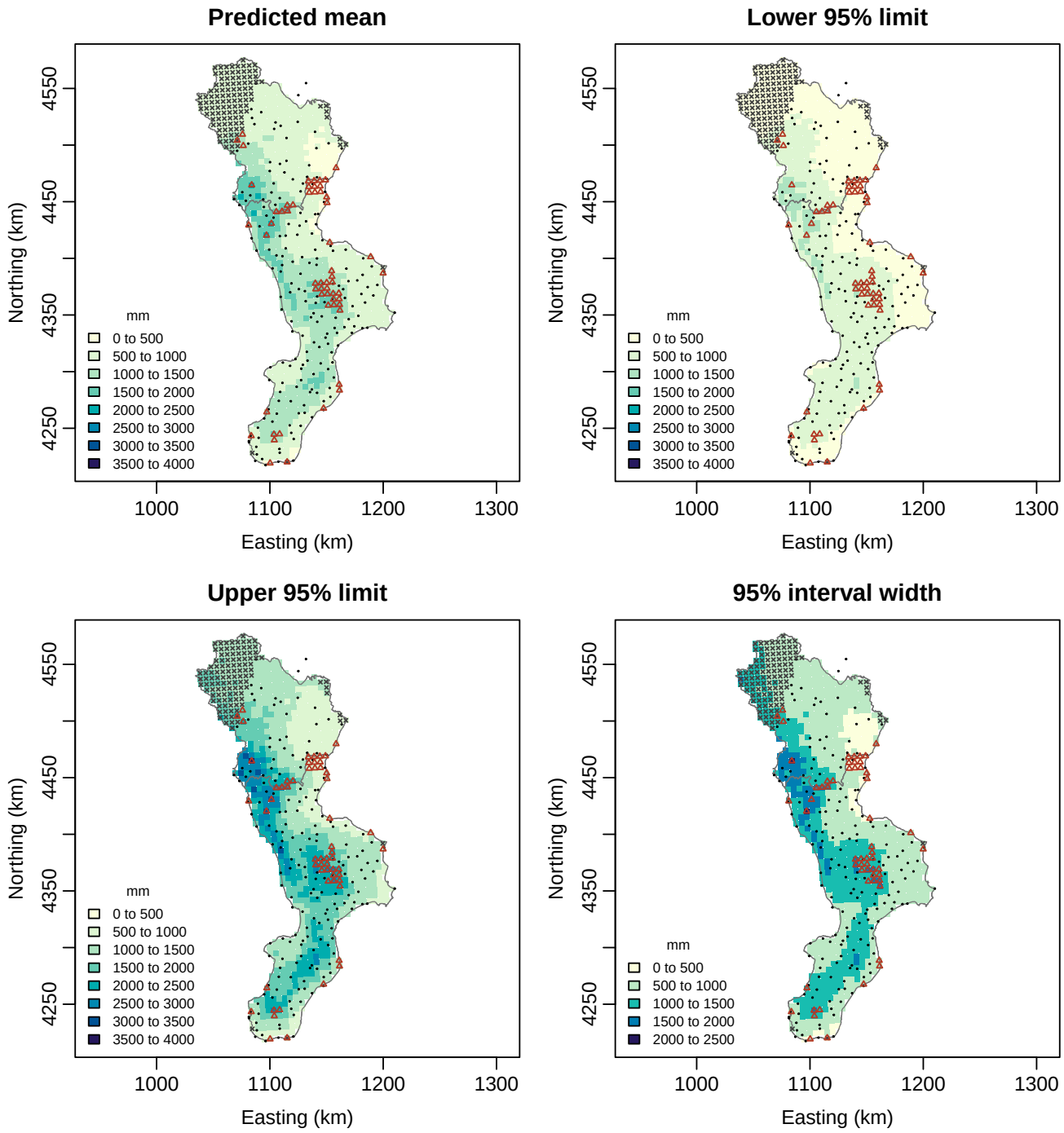
I use model 4: Log, Elevation_xy, exponential. Selection interval score is 1382.2 mm and RMSE is 291.9 mm; held-out coverage is 95.7%. Predicted means range from 495.8 to 2896.7 mm. The median interval width is 936.6 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.5 Year 1974

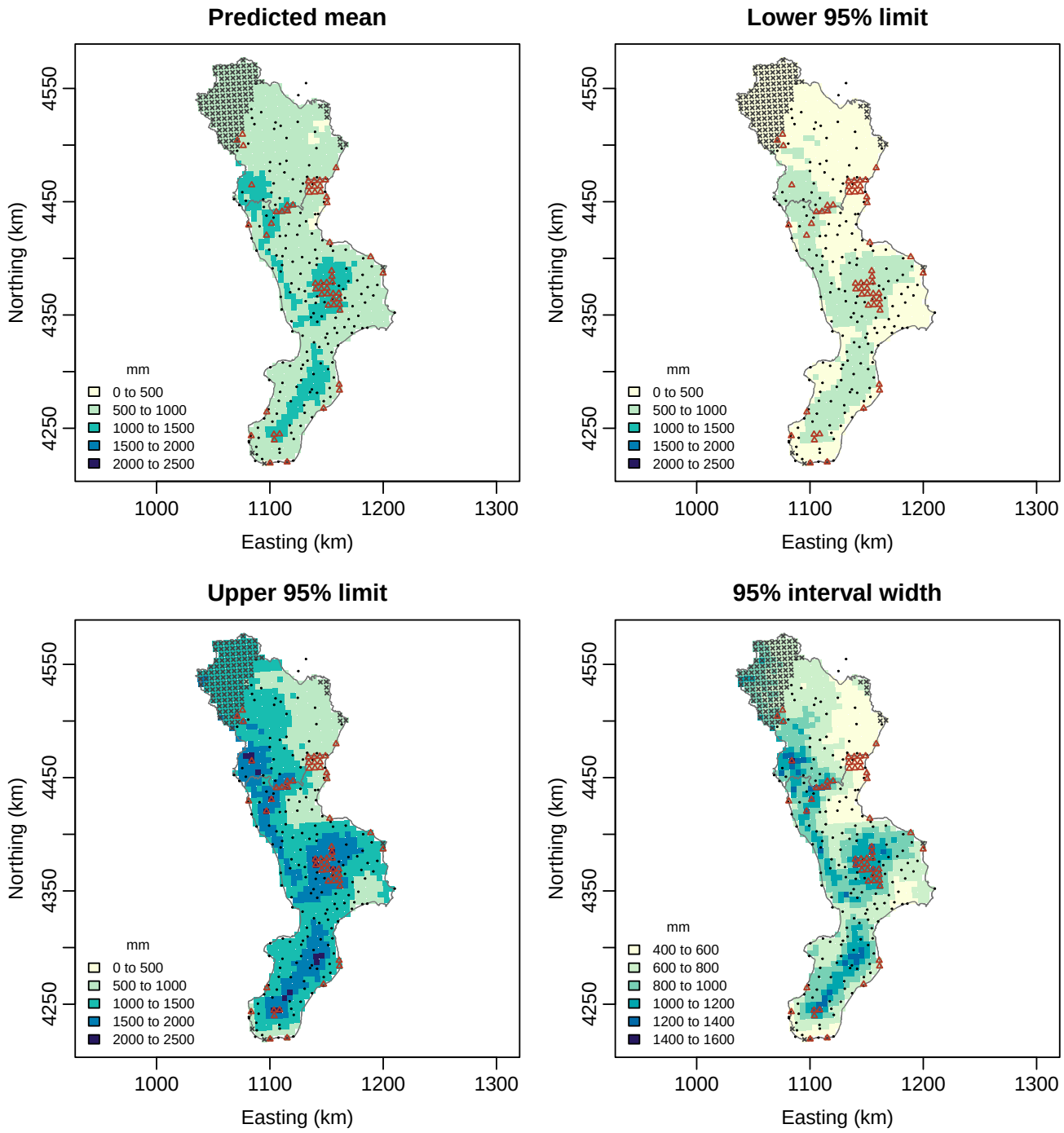
I use model 10: Log, Elevation, matern. Selection interval score is 1233.8 mm and RMSE is 255.6 mm; held-out coverage is 95.7%. Predicted means range from 408.5 to 2289.8 mm. The median interval width is 883.4 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.6 Year 1975

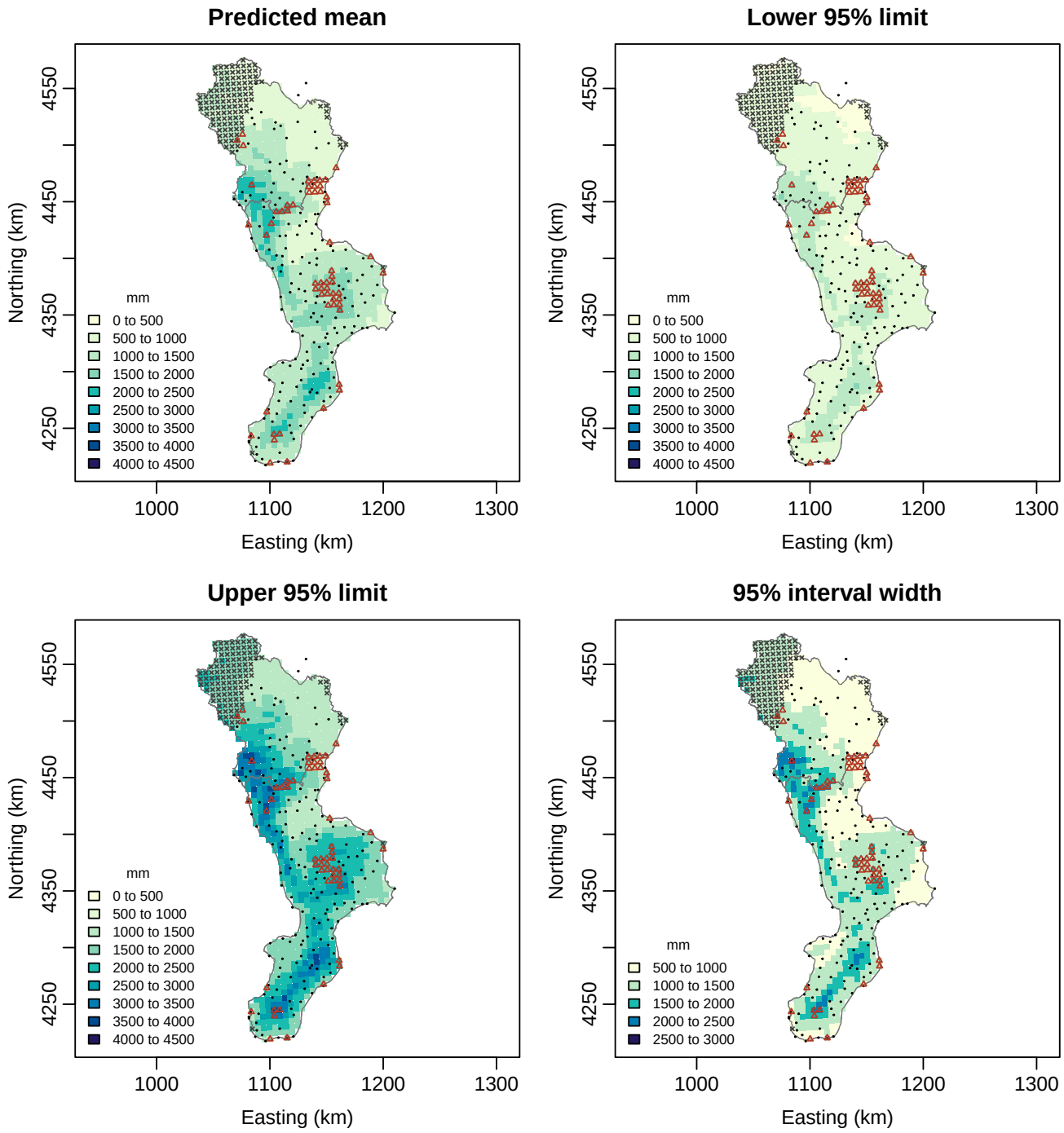
I use model 10: Log, Elevation, matern. Selection interval score is 1003.1 mm and RMSE is 192.9 mm; held-out coverage is 92.9%. Predicted means range from 452.4 to 1495.2 mm. The median interval width is 778.3 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 119 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.7 Year 1976

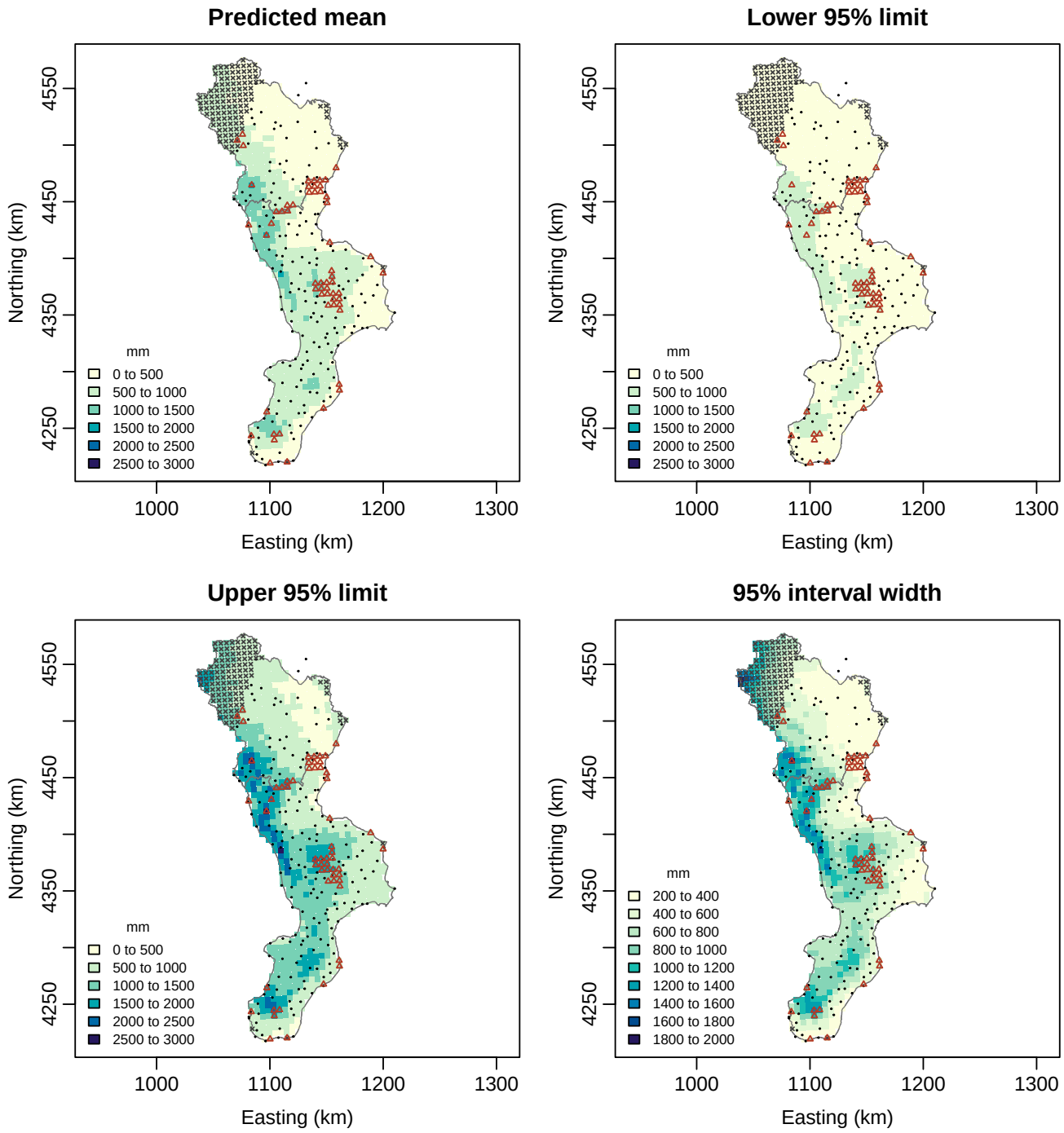
I use model 2: Log, Elevation, exponential. Selection interval score is 1576.3 mm and RMSE is 334.5 mm; held-out coverage is 95.0%. Predicted means range from 726.7 to 2725.0 mm. The median interval width is 1143.9 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.8 Year 1977

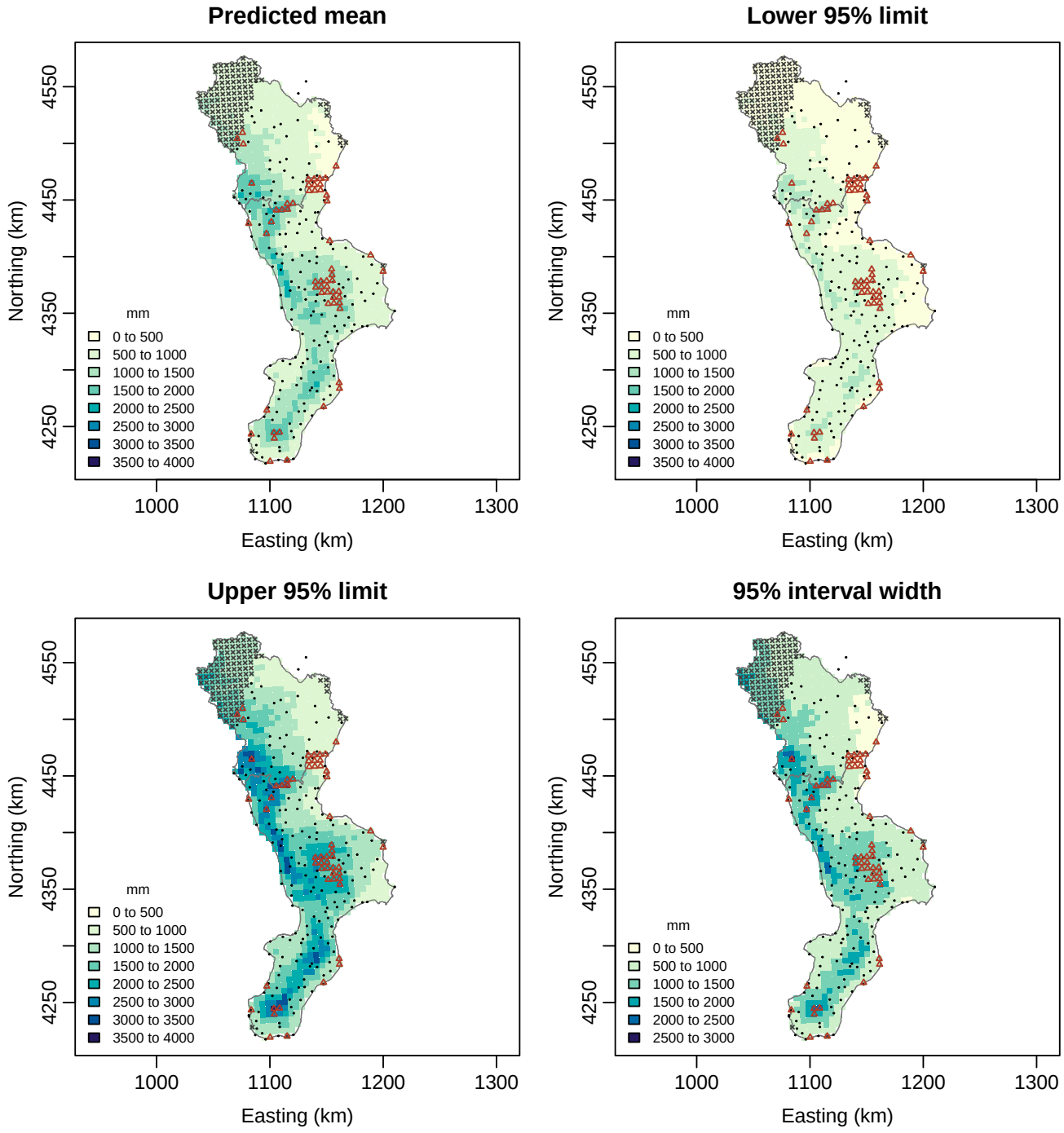
I use model 12: Log, Elevation_xy, matern. Selection interval score is 908.8 mm and RMSE is 182.3 mm; held-out coverage is 92.9%. Predicted means range from 237.9 to 1579.6 mm. The median interval width is 649.9 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.9 Year 1978

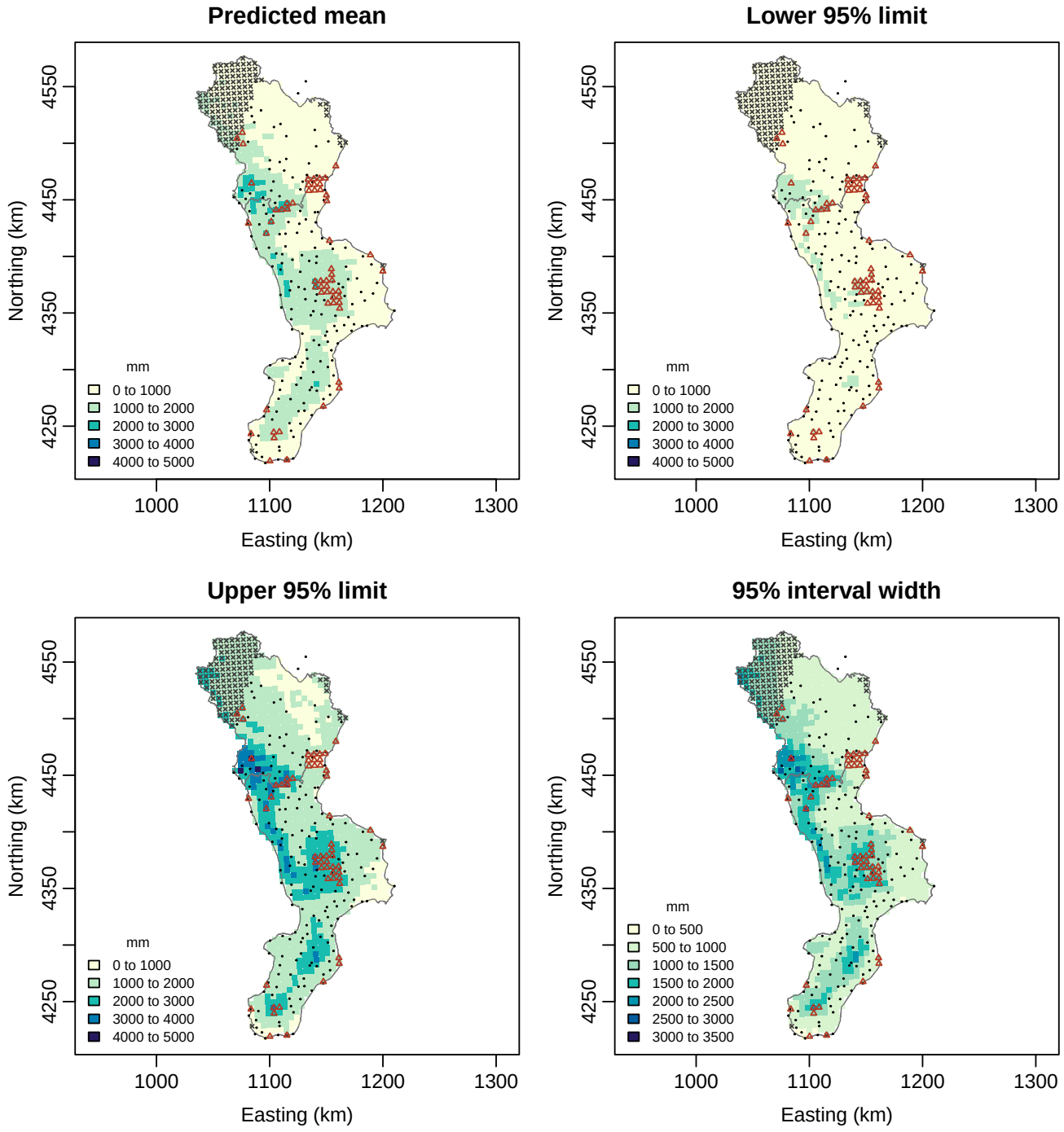
I use model 4: Log, Elevation_xy, exponential. Selection interval score is 1355.0 mm and RMSE is 253.3 mm; held-out coverage is 95.1%. Predicted means range from 414.1 to 2464.7 mm. The median interval width is 934.2 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.10 Year 1979

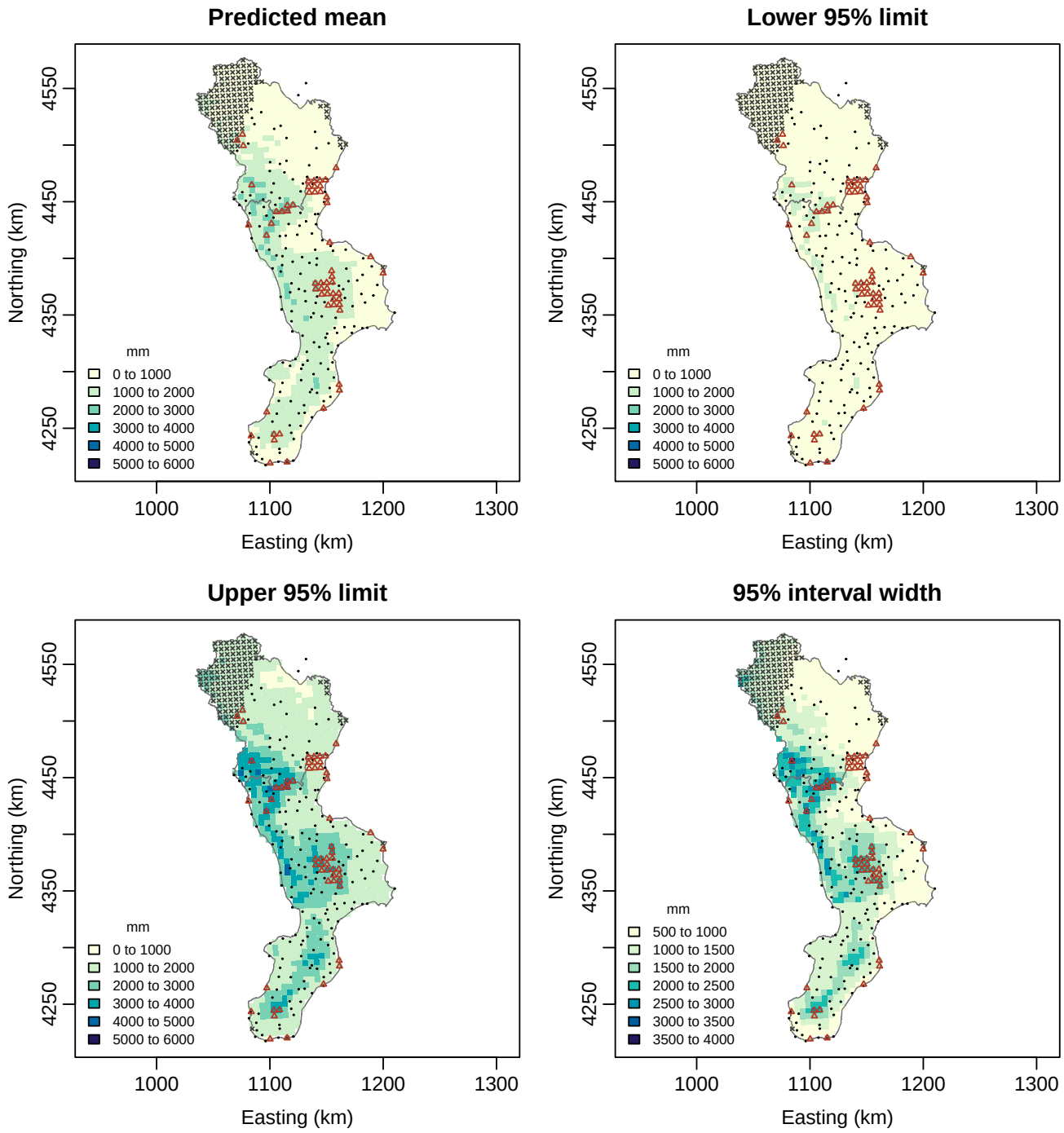
I use model 4: Log, Elevation_xy, exponential. Selection interval score is 1422.8 mm and RMSE is 266.2 mm; held-out coverage is 93.9%. Predicted means range from 393.8 to 2849.6 mm. The median interval width is 986.5 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.11 Year 1980

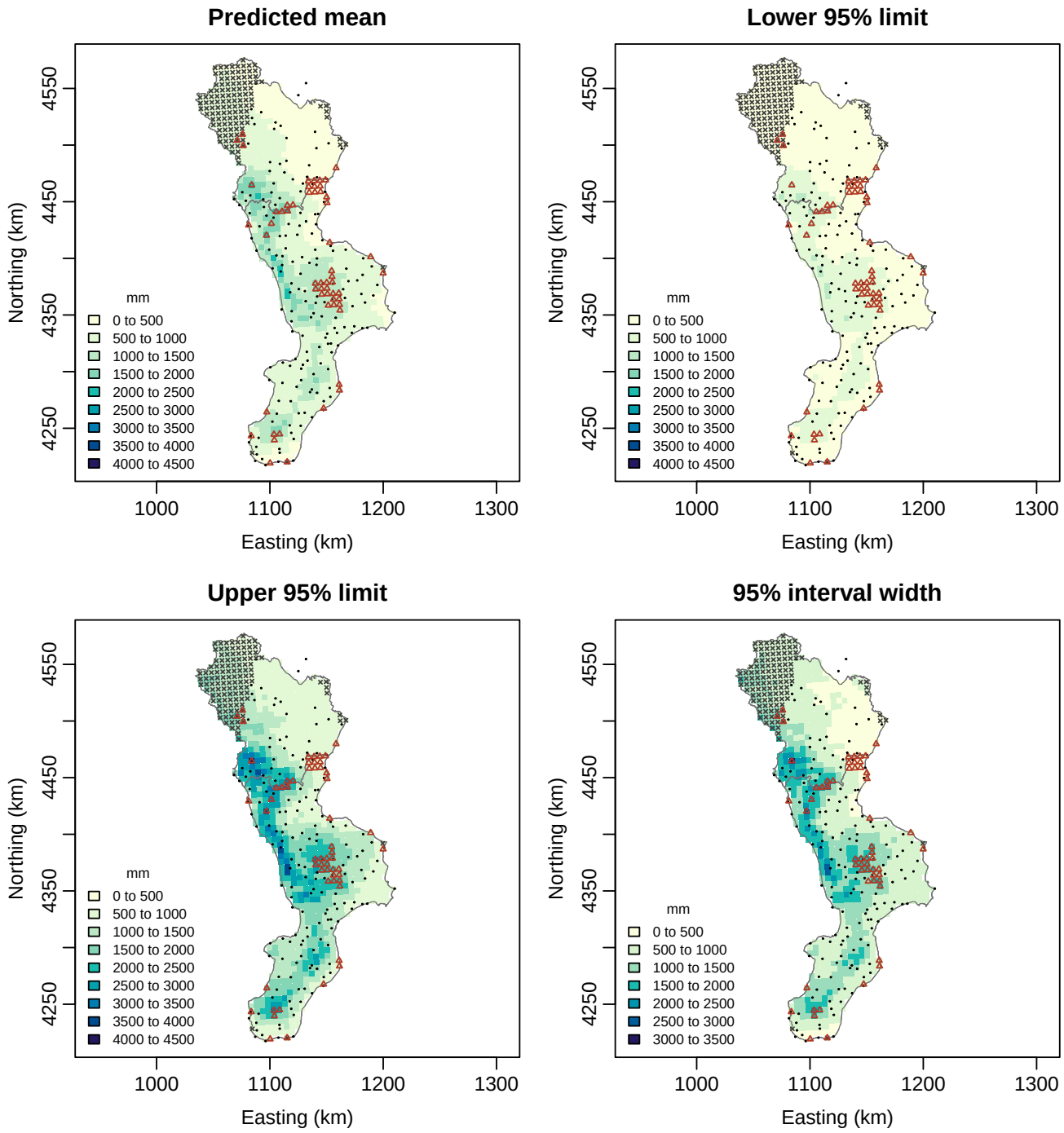
I use model 8: Log, Elevation_xy, spherical. Selection interval score is 1829.4 mm and RMSE is 333.7 mm; held-out coverage is 94.9%. Predicted means range from 490.4 to 2966.2 mm. The median interval width is 1203.5 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 117 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.12 Year 1981

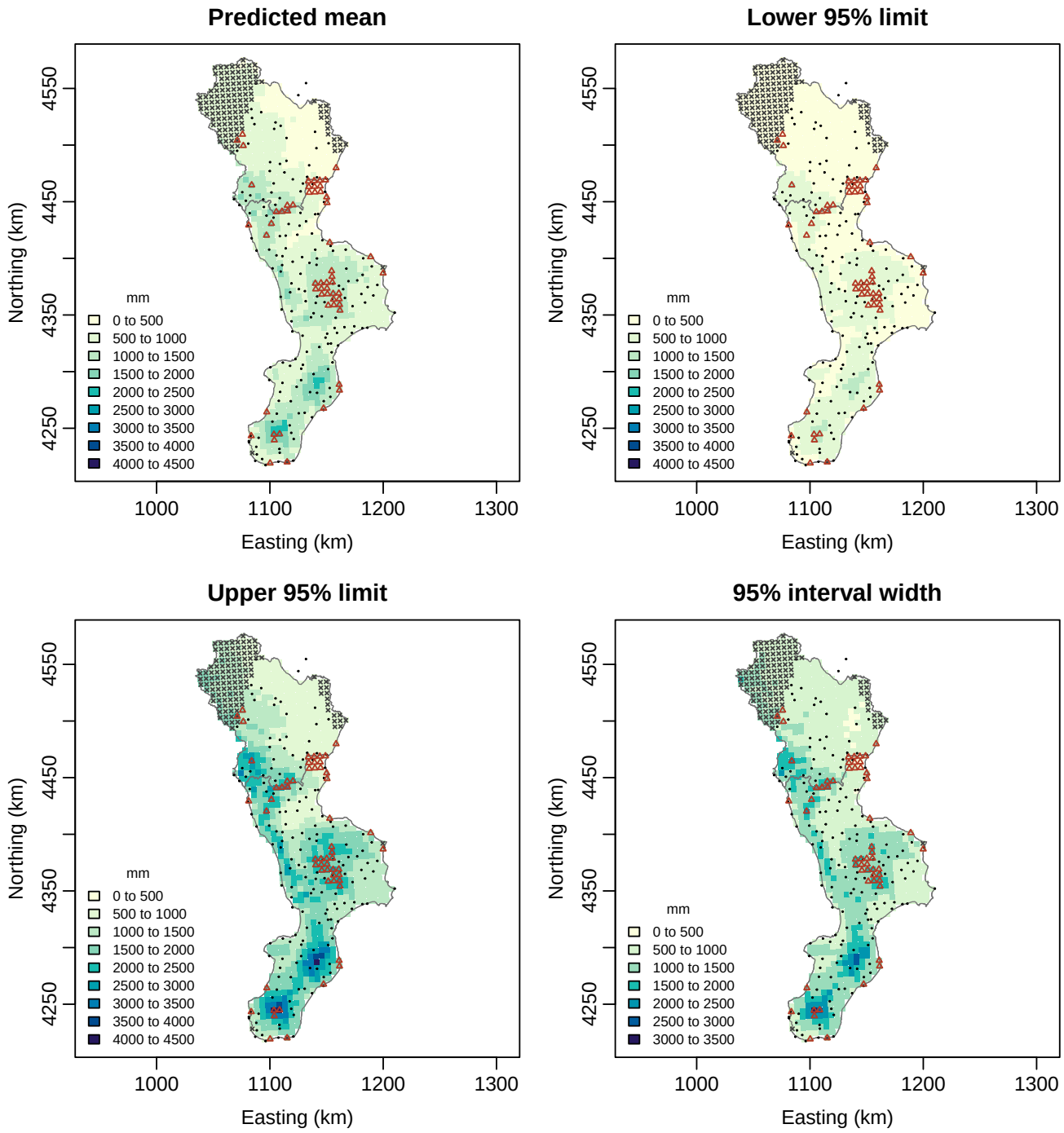
I use model 2: Log, Elevation, exponential. Selection interval score is 1340.3 mm and RMSE is 267.4 mm; held-out coverage is 94.2%. Predicted means range from 286.3 to 2380.1 mm. The median interval width is 889.0 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 126 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

6.13 Year 1982

I use model 2: Log, Elevation, exponential. Selection interval score is 1316.5 mm and RMSE is 243.5 mm; held-out coverage is 94.2%. Predicted means range from 387.3 to 2456.2 mm. The median interval width is 976.8 mm.



Crosses: outside station hull. Red triangles: elevation outside the observed range. 127 grid points are outside the hull and 57 have an extrapolated elevation. These groups may overlap.

7 What I take from the results

Across the maps, higher predicted rainfall often follows the higher elevations, with local variation added by the spatial residual correction. This is consistent with the positive rainfall–elevation association seen in the observations. It does not mean elevation explains every wet or dry location.

The maps separate the estimated rainfall pattern from its uncertainty. I should read the predicted mean together with the lower and upper limits, especially near the boundary and at elevations beyond the station range. A detailed-looking map is not evidence that every local feature is well known.

The best-scoring model is allowed to change by year. This does not mean that the physical process suddenly changes covariance family each year: finite samples, changing stations and small score differences can also change the ranking. I use one common set of candidate models and one common selection rule to make the yearly analysis consistent.

The important limits are the assumed trend and residual stationarity, isotropy, Gaussian approximation on the selected scale, and treating the estimated covariance parameters as known. The validation split also matters. Repeating the splits or using spatially separated validation would help assess stability, while a bootstrap or a Bayesian analysis could address more parameter uncertainty. Those are extensions, not analyses performed here.

8 Appendix: all candidate scores

The tables below make the selection rule checkable. `IS95` and `Width` are in mm, `Coverage` is a percentage, and the Matérn model always has $\kappa = 1.5$. Rows are ordered by interval score. The first row is selected.

```
for (r in results) {
  cat("\n\n### ", r$year, "\n\n", sep = "")
  tab <- r$scores[order(r$scores$IS95, r$scores$RMSE),
                  c("Model", "IS95", "RMSE", "Coverage", "Width")]
  print(knitr::kable(tab, digits = 2, row.names = FALSE))
  cat("\n\n")
}
```

8.0.1 1970

Model	IS95	RMSE	Coverage	Width
6	1009.55	195.42	92.59	663.77
10	1015.56	195.84	93.12	664.71
12	1016.90	196.33	93.12	666.54
4	1020.25	196.16	92.59	666.76
2	1022.67	195.59	92.59	665.03
8	1035.17	198.16	92.59	667.94
7	1312.73	204.40	92.06	758.56
1	1315.89	201.02	93.12	754.92
3	1316.60	201.87	93.12	756.82
9	1323.15	201.38	92.59	756.12
11	1326.96	202.15	92.59	758.09
5	1329.29	204.88	92.06	755.03

8.0.2 1971

Model	IS95	RMSE	Coverage	Width
6	1020.53	204.66	91.53	665.68
2	1023.84	205.56	92.06	668.61
12	1026.96	209.42	93.12	673.97
8	1028.28	205.46	91.53	665.66

Model	IS95	RMSE	Coverage	Width
10	1031.44	209.95	92.59	674.54
4	1032.48	206.31	92.06	669.60
9	1289.18	205.30	95.24	776.36
5	1289.49	202.45	94.18	773.54
11	1296.28	204.88	94.71	776.24
1	1301.70	202.33	94.18	774.65
3	1308.33	203.15	94.71	776.13
7	1311.26	206.09	94.71	775.45

8.0.3 1972

Model	IS95	RMSE	Coverage	Width
6	1312.81	257.13	94.65	973.76
10	1322.91	258.75	93.58	979.36
8	1334.43	259.52	94.65	976.64
2	1334.81	259.02	94.12	982.47
12	1341.25	260.25	94.12	983.52
4	1350.70	260.81	94.12	987.29
5	1434.53	255.67	93.58	968.46
7	1447.71	258.19	93.05	969.65
9	1455.32	257.04	93.58	975.05
11	1470.45	259.21	93.58	979.19
1	1476.75	258.74	93.58	984.48
3	1487.42	260.96	93.58	989.20

8.0.4 1973

Model	IS95	RMSE	Coverage	Width
4	1382.17	291.85	95.74	1103.43
2	1383.84	291.58	95.74	1100.65
12	1401.32	295.66	94.15	1109.48
6	1402.78	292.44	94.68	1097.78
10	1418.60	296.78	94.15	1107.36
8	1423.55	296.67	95.21	1099.92
1	1486.95	288.11	94.68	1112.05
3	1487.58	287.93	94.15	1115.77
7	1492.48	292.95	93.62	1109.98
11	1506.50	290.84	94.68	1115.41
9	1517.53	293.31	92.55	1113.05
5	1532.76	295.98	93.62	1111.23

8.0.5 1974

Model	IS95	RMSE	Coverage	Width
10	1233.84	255.59	95.72	954.96
12	1235.48	256.11	95.19	956.38
4	1287.20	253.21	94.65	967.66
8	1287.44	254.67	94.65	963.61
2	1289.03	252.72	95.19	967.78
6	1291.78	254.06	94.65	963.54

Model	IS95	RMSE	Coverage	Width
9	1497.31	258.86	93.05	972.51
11	1504.70	258.99	93.05	974.74
7	1505.61	256.84	93.58	965.78
5	1513.85	258.67	93.05	964.50
1	1530.64	256.35	93.05	973.07
3	1533.43	256.95	93.05	975.21

8.0.6 1975

Model	IS95	RMSE	Coverage	Width
10	1003.12	192.90	92.86	798.05
2	1008.13	191.76	92.86	803.17
6	1009.86	193.34	92.31	799.92
1	1018.50	191.62	92.31	746.18
8	1019.18	193.68	92.31	800.05
12	1019.54	193.54	92.86	800.87
5	1022.82	193.71	92.31	744.08
4	1022.94	192.75	92.86	806.39
3	1027.56	192.63	91.21	749.43
9	1035.34	192.45	92.31	745.17
11	1040.25	193.12	91.76	747.89
7	1040.65	193.13	91.21	745.62

8.0.7 1976

Model	IS95	RMSE	Coverage	Width
2	1576.34	334.48	95.03	1235.90
4	1577.56	335.12	95.03	1239.39
6	1586.86	337.08	95.03	1230.44
8	1599.27	338.93	95.03	1232.47
10	1600.20	336.49	94.48	1234.71
12	1603.92	336.84	94.48	1238.14
1	1877.04	337.74	92.27	1328.29
3	1881.78	338.54	92.27	1332.96
9	1902.97	340.84	92.82	1322.87
5	1907.43	342.82	92.82	1320.63
11	1908.82	340.95	92.82	1327.24
7	1926.51	344.04	92.82	1324.11

8.0.8 1977

Model	IS95	RMSE	Coverage	Width
12	908.78	182.30	92.90	668.21
10	908.90	181.70	92.90	666.14
4	920.62	183.12	92.90	677.91
8	922.02	184.14	92.35	675.46
2	922.37	182.55	92.35	676.81
6	922.96	183.09	92.35	674.59
3	1118.07	186.96	94.54	724.04
1	1119.20	185.67	94.54	722.12

Model	IS95	RMSE	Coverage	Width
5	1123.49	187.01	94.54	718.10
7	1127.43	187.97	94.54	719.46
11	1131.66	186.35	94.54	721.14
9	1134.03	185.24	94.54	718.59

8.0.9 1978

Model	IS95	RMSE	Coverage	Width
4	1355.00	253.33	95.05	1022.90
2	1356.69	251.21	95.60	1017.93
6	1368.80	252.03	94.51	1017.04
8	1374.02	257.94	94.51	1019.61
12	1376.90	256.87	94.51	1021.67
10	1396.00	256.54	95.05	1017.74
3	1427.58	255.80	94.51	1036.66
11	1429.94	258.50	95.05	1038.01
1	1433.63	252.34	94.51	1034.24
7	1445.83	259.60	93.41	1034.28
9	1463.03	256.13	92.86	1037.87
5	1468.55	258.24	93.41	1033.40

8.0.10 1979

Model	IS95	RMSE	Coverage	Width
4	1422.80	266.25	93.92	1083.03
2	1423.78	265.22	93.92	1082.82
6	1430.51	267.16	94.48	1081.54
8	1430.54	267.43	93.92	1079.74
12	1439.09	271.87	93.92	1083.98
10	1439.53	270.93	93.92	1083.80
1	1577.96	258.37	93.37	1070.49
3	1579.61	260.70	93.37	1073.19
5	1581.23	261.91	93.92	1070.89
7	1583.17	264.64	93.37	1074.62
9	1598.12	262.28	93.37	1073.71
11	1598.56	264.30	93.37	1076.43

8.0.11 1980

Model	IS95	RMSE	Coverage	Width
8	1829.40	333.74	94.94	1346.19
10	1833.04	334.37	94.94	1350.80
12	1838.05	334.32	94.94	1352.45
2	1842.92	333.53	94.38	1346.84
4	1843.11	333.44	94.38	1348.46
6	1854.39	333.96	94.94	1347.48
3	2097.55	330.86	92.13	1323.09
11	2098.86	331.12	92.13	1328.83
1	2098.91	330.01	92.13	1320.72
9	2102.14	329.98	91.57	1327.30

Model	IS95	RMSE	Coverage	Width
5	2117.74	330.64	91.57	1325.52
7	2135.15	336.07	91.57	1324.68

8.0.12 1981

Model	IS95	RMSE	Coverage	Width
2	1340.30	267.35	94.22	1019.16
4	1342.66	267.99	94.80	1020.37
6	1353.95	268.51	94.22	1018.71
8	1364.70	270.15	94.22	1021.09
12	1401.42	271.17	94.22	1017.07
10	1404.91	271.00	94.22	1014.95
3	1608.74	268.03	93.64	1056.68
1	1612.91	266.73	93.64	1054.05
7	1637.78	270.10	93.64	1054.96
5	1647.23	270.37	93.64	1051.15
11	1649.54	271.78	93.64	1059.96
9	1655.01	270.54	93.64	1057.62

8.0.13 1982

Model	IS95	RMSE	Coverage	Width
2	1316.53	243.52	94.22	1058.31
6	1319.31	245.44	94.22	1058.25
12	1321.83	248.67	94.22	1058.34
10	1323.15	247.25	94.80	1061.27
4	1325.18	245.76	94.80	1062.17
8	1329.55	250.06	93.64	1057.61
11	1361.31	240.67	93.06	933.17
7	1364.54	242.64	91.91	929.36
3	1365.39	239.79	92.49	939.23
1	1372.29	239.88	92.49	935.70
9	1380.80	243.16	92.49	935.59
5	1395.78	243.19	91.91	937.71

9 Sources and reproducibility

I used the supplied course notes by Giovanna Jona Lasinio, *Chapter1-2* (exploration, trend and variograms), *Chapter3-5* (Gaussian spatial models, maximum likelihood and kriging uncertainty), and *Chapter6-11*, Section 10.3 (predictive validation and interval score). The last score applies to central prediction intervals here; I am not using Bayesian estimation. I also checked the `geoR` definitions of `likfit`, `trend.spatial`, `krige.control` and `krige.conv`, especially external covariates, Matérn smoothness and the observation prediction target.

```
packages <- c("geoR", "sp", "knitr", "rmarkdown")
knitr::kable(data.frame(Component = c("R", packages),
  Version = c(as.character(getRversion()),
    vapply(packages, function(p) as.character(packageVersion(p)),
           character(1)))), row.names = FALSE)
```

Component	Version
R	4.5.0
geoR	1.9.6
sp	2.2.0
knitr	1.50
rmarkdown	2.29

10 Use of AI

I used ChatGPT/Codex to help develop the analysis plan, write and run the R code, check the prediction and interval calculations, and draft the comments in this report. AI assistance covered the data checks, yearly model comparisons, likelihood fits, validation, maps and compilation. I asked for simple code and explanations that I could follow and discuss in class. The main request was to analyse every rainfall year using the matching grid, choose trend and covariance models by predictive scores, and report pointwise intervals with their assumptions. The computations were run in the AI-assisted R environment. This report does not claim an independent manual replication by me; I remain responsible for reviewing the results and understanding the choices before presenting them.